Pricing Prediction Markets: Incomplete Markets, Selection Rules, and Risk Premia

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Abstract

Prediction market contracts are binary contingent claims written on real-world events whose payoffs are generally not spanned by traded assets. In such incomplete markets, no-arbitrage determines a set of admissible pricing measures rather than a unique price, so observed contract prices need not equal physical event probabilities. This paper proposes a structural selection framework for prediction markets. The event is generated by a latent Gaussian threshold factor under the physical measure, and the market-selected pricing measure is modeled as a one-parameter exponential tilt of the standardized latent factor. Under this maintained Gaussian family, the observed market price satisfies $p_t^{\mathrm{mkt}} = \Phi(\Phi^{-1}(p_t^*) + \lambda_t)$, so the Wang transform arises exactly as a variance-preserving pricing-measure selection rule. The model implies favorite-longshot bias as a corollary and nests a broader probit-linear family when conditional variance is not preserved. I estimate the selection parameter from resolved prediction market contracts and find a positive and statistically significant wedge on real-money platforms, with larger values in lower-liquidity and longer-duration markets and a monotone decline over contract life. Across 291,309 contracts from multiple platforms, the estimated parameter is positive on real-money venues and negative on the play-money platform Manifold. These results indicate a systematic price-probability wedge under the maintained Wang family: on real-money venues, observed prices lie above extracted physical probabilities on average, while the play-money platform Manifold exhibits the opposite sign. Under complementarity, the implied cross-sectional price distribution is compressed toward 0.5. Within incomplete event markets, the maintained Wang family provides a tractable structural benchmark for pricing and probability extraction.

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1 Introduction

Prediction market prices are widely interpreted as probabilities. However, prediction market contracts are binary contingent claims whose payoffs are typically not spanned by traded assets. In such incomplete markets, no-arbitrage determines a *set* of admissible pricing measures rather than a unique one, so the observed market price need not equal the physical event probability (Manski, 2006; Harrison and Pliska, 1983). This interpretation problem has practical consequences: without a pricing model, there is no principled way to extract physical probabilities from market prices, to compare contracts on a risk-adjusted basis, or to explain systematic biases such as the favorite-longshot bias documented across 300,000+ Kalshi contracts by Bürgi, Deng, and Whelan (2025).

This paper proposes a structural selection framework that addresses these gaps. The event is generated by a latent Gaussian threshold factor under the physical measure. Within the incomplete-market set of admissible pricing measures, I impose a maintained one-parameter exponential tilt of the standardized latent factor. Under this family, the Wang (2000) transform arises exactly:

$$p_t^{\mathrm{mkt}} = \Phi\Big(\Phi^{-1}(p_t^*) + \lambda\Big)\,,$$

where λ is the pricing-measure selection parameter. Favorite-longshot bias emerges as a corollary, and a broader probit-linear family nests the Wang restriction as the variance-preserving special case. This claim is conditional on the maintained Gaussian family; it is not implied by no-arbitrage alone.

Empirically, I estimate the implied selection parameter from resolved prediction market contracts and find a positive, statistically significant wedge on real-money platforms, with larger values in lower-liquidity and longer-duration markets and a monotone decline over contract life. Across 291,309 contracts from Polymarket, Kalshi, Metaculus, Good Judgment Open, INFER, and Manifold, the estimated parameter is positive on all real-money venues and negative on the play-money platform Manifold—patterns consistent with the interpretation that risk-bearing costs are more important on real-money platforms, while confidence-related distortions dominate when financial stakes are absent.

The framework provides practitioners with a tractable benchmark for extracting physical probabilities, comparing contracts across platforms, and organizing the pricing wedge's cross-sectional and temporal variation.

Contributions. This paper makes four contributions.

First, it gives a formal incomplete-market foundation for prediction market pricing. When the event payoff is not spanned by traded claims, no-arbitrage leaves a non-singleton set of admissible pricing measures rather than a unique price (Section 3.1).

Second, it proposes a structural selection rule within that incomplete-market set. Under a latent Gaussian threshold factor and a one-parameter exponential tilt of the standardized latent

factor, the Wang transform arises exactly. A broader probit-linear family emerges when variance preservation is relaxed (Section 3.3).

Third, it derives favorite-longshot bias as a corollary of the maintained pricing rule and interprets the selection parameter economically as

$$\lambda_t = -\beta_t \theta_t / \Sigma_{t,T},$$

where β_t is factor exposure, θ_t is the market price of the priced factor, and $\Sigma_{t,T}$ is total event uncertainty (Section 3.6).

Fourth, it estimates the implied selection parameter in a large sample of resolved contracts and documents systematic heterogeneity by liquidity, duration, and platform type, including a negative estimate on the play-money platform Manifold (Sections 5–6).

The remainder of the paper is organized as follows. Section 2 reviews related literature. Section 3 develops the theoretical framework: incompleteness, the latent threshold factor, the Wang transform as a maintained selection rule, and the complete pricing model. Section 4 describes data and sample construction. Section 5 presents the main empirical results. Section 6 provides cross-platform validation. Section 7 discusses extensions. Section 8 concludes.

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2 Related Literature

This paper sits at the intersection of four literatures: stochastic modeling of prediction markets, the interpretation of prediction market prices, the favorite-longshot bias, and applications of the Wang transform.

Stochastic modeling. Dalen (2025) proposes a logit jump-diffusion in risk-neutral ($\mathbb{Q}$) space with a martingale drift constraint, calibration pipeline, and belief-volatility surface. Dalen's contribution and mine are complementary but conceptually distinct. Dalen takes an admissible pricing measure as given and develops an observation-space dynamics for market making in logit space. This paper instead asks which admissible pricing measure within an incomplete-market set best organizes observed prices and how that maintained pricing family relates to the physical measure P. The Wang specification is not presented as a generic P-to-$\mathbb{Q}$ bridge for arbitrary dynamics, but as an exact selection rule under the latent Gaussian threshold model developed in Section 3. In that sense, Dalen provides a useful empirical dynamics in quote space, while this paper provides a maintained pricing-measure selection rule for incomplete event markets. Earlier, Archak and Ipeirotis (2009) modeled InTrade prices using Itô diffusions for latent ability processes, deriving contract price volatility as a function of current price and time to expiry.

Probability interpretation. Manski (2006) showed that interpreting prediction market prices directly as probabilities lacks general formal justification: equilibrium price reflects a functional

of heterogeneous beliefs and need not equal the mean belief. Wolfers and Zitzewitz (2006) showed that under log utility and related specifications, prices can approximate mean beliefs; Gjerstad (2004) proved the sufficiency of the log-utility condition. This paper does not attempt a nonparametric recovery of beliefs from prices alone. Instead, it imposes a maintained pricing family within the incomplete-market admissible set and uses that family to map between physical probabilities and observed prices. In this sense, the paper operationalizes the Manski critique through a structural selection rule rather than through a generic calibration map.

Favorite-longshot bias. The FLB has been documented in horse racing (Griffith, 1949; Snowberg and Wolfers, 2010), sports betting (Whelan, 2024), and prediction markets (Bürgi, Deng, and Whelan, 2025). Explanations include risk-loving preferences (Ali, 1977), probability weighting (Barberis and Huang, 2008), adverse selection (Shin, 1991, 1993), and noise (Ottaviani and Sorensen, 2010). Jullien and Salanié (2000) estimated expected utility, rank-dependent, and cumulative prospect theory models on racetrack data, finding that inverse-S shaped probability weighting—overweighting of small probabilities and underweighting of large ones—was a key driver of FLB, with cumulative prospect theory providing the best fit. The Wang transform provides a parametric complement to these nonparametric estimates. The Wang transform provides a unifying structural mechanism that nests the risk-averse market maker explanation (Whelan, 2024).

Wang transform applications. Wang (2000, 2004) applied the transform to catastrophe bond pricing ($\hat{\lambda} \approx 0.45$) and proposed it as a universal pricing framework for financial and insurance risks. Kijima and Muromachi (2008) derived the transform from Bühlmann's economic premium principle, establishing an equilibrium foundation. Hamada and Sherris (2003) showed it recovers the Black-Scholes formula in the lognormal case and related the distortion parameter to systematic risk. Pelsser (2008) showed that the Wang transform is not consistent with arbitrage-free pricing for general stochastic processes, implying it is not a universal pricing framework; the present application to incomplete markets sidesteps this critique. Huang, Sun, and Zhang (2024) recently generalized the transform to flexible probability weighting functions for lottery-like equities, embedding it within the Pi-CAPM framework. To the best of my knowledge, this is the first paper to apply the Wang transform to prediction markets and to derive favorite-longshot bias as a corollary of that maintained pricing rule.

Prediction market design and calibration. Wolfers and Zitzewitz (2004) provide the standard survey on prediction market contract design and information aggregation. Servan-Schreiber, Wolfers, Pennock, and Galebach (2004) compare real-money and play-money prediction markets, finding similar forecasting accuracy—a result relevant to the interpretation of the Manifold play-money control in Section 6. Le (2026) argues that prediction market calibration is structured by horizon, domain, and trade size, complementing the heterogeneity analysis in Sections 5.3 and 5.6. Ostrovsky (2012) proves that in dynamic trading with separable securities, prices converge to the pooled-information posterior—a result relevant to the time-varying λ decay documented in Section 5.4. Gneiting, Balabdaoui, and Raftery (2007) provide the foundational treatment of probabilistic calibration, against which the Wang transform can be

understood as a parametric recalibration map.

Prediction market returns and equilibrium. Within the *Quantitative Finance* literature, Bottazzi and Giachini (2019) study wealth-driven selection in prediction markets, showing that beliefs closer to the truth survive in the long run but that convergence can be slow and non-monotone—a result relevant to the persistence of the pricing wedge documented here. Whelan (2023) derives analytic expressions for expected returns in binary prediction markets and shows that risk premia arise naturally when traders are risk-averse, providing an equilibrium foundation consistent with the positive $\hat{\lambda}$ estimated in Sections 5–6.

Bayesian approaches. Madrigal-Cianci, Monsalve Maya, and Breakey (2026) recast prediction markets as Bayesian inverse problems in log-odds space, providing posterior uncertainty quantification on probability estimates.

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3 Theoretical Framework

3.1 Prediction Markets as Incomplete-Market Binary Options

A prediction market contract pays \$1 if event E occurs at or before time T, and \$0 otherwise. This is the payoff of a cash-or-nothing binary option (Rubinstein and Reiner, 1991):

$$\text{Payoff} = \$1 \cdot \mathbf{1}_{\{E \text{ occurs}\}}$$

In standard options theory, the Black-Scholes price of a cash-or-nothing binary call is:

$$C_{\text{binary}} = e^{-rT}\,\Phi(d_2), \quad d_2 = \frac{\ln(S/K) + (r - \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$$

The contract price p_t is the risk-neutral probability of the event:

$$p_t = e^{-r\tau}\,\mathbb{E}^{\mathbb{Q}}[\mathbf{1}_{\{E\}} \mid \mathcal{F}_t]$$

For the short maturities typical of prediction markets (days to months), $e^{-r\tau} \approx 1$, so $p_t \approx \mathbb{E}^{\mathbb{Q}}[\mathbf{1}_E \mid \mathcal{F}_t]$. The structural equivalence tells us *what* to price. But standard tools cannot tell us *how*, because the market is incomplete.

Terminology. Throughout this paper, I use the following terms consistently. The *physical probability* p^* denotes the objective likelihood of event E under the data-generating measure P. The *market price* p^{mkt} denotes the observed contract price. The *risk-adjusted probability* $\hat{p}^* = \Phi(\Phi^{-1}(p^{\text{mkt}}) - \hat{\lambda})$ denotes the estimated physical probability after stripping the pricing

wedge. I avoid the terms `true probability,''`fair value,'' and "actuarially fair value'' except where context requires them, as they can conflate distinct concepts.

For precision, I state the incompleteness argument in a discounted finite-state model; the economic conclusion is the standard one: if the event payoff is not spanned by traded claims, no-arbitrage leaves an interval of admissible prices rather than a unique price.

Proposition 1 (Unattainable event claims imply incompleteness and a price interval)**.** *Work in discounted units on a finite terminal state space*

$$\Omega = \{\omega_1, \ldots, \omega_n\}.$$

Let $\mathcal{M}_T \subset \mathbb{R}^n$ denote the linear space of all terminal discounted claims attainable by admissible self-financing strategies with arbitrary initial capital. Let $\pi_0 : \mathcal{M}_T \to \mathbb{R}$ be the time-0 pricing functional on attainable claims. Assume the discounted market is arbitrage-free, so that the set of equivalent martingale measures

$$\mathcal{Q} := \left\{ q \in \Delta^{n-1}_{++} : \pi_0(X) = E_q[X] \text{ for all } X \in \mathcal{M}_T \right\}$$

is nonempty, where

$$\Delta^{n-1}_{++} := \left\{ q \in \mathbb{R}^n_{++} : \sum_{i=1}^n q_i = 1 \right\}.$$

Let

$$H := \mathbf{1}_E \in \mathbb{R}^n$$

be the discounted payoff of the binary event claim. If

$$H \notin \mathcal{M}_T,$$

then:

1. *H is unattainable.*
2. *The market is incomplete.*
3. *$\mathcal{Q}$ contains at least two distinct equivalent martingale measures.*
4. *The set of arbitrage-free prices for H is*

$$\Pi(H) := \{E_q[H] : q \in \mathcal{Q}\},$$

and $\Pi(H)$ is a nondegenerate compact interval:

$$\Pi(H) = [\underline{\pi}(H), \overline{\pi}(H)], \qquad \underline{\pi}(H) < \overline{\pi}(H),$$

with

$$\underline{\pi}(H) := \min_{q \in \mathcal{Q}} E_q[H], \qquad \overline{\pi}(H) := \max_{q \in \mathcal{Q}} E_q[H].$$

Proof. Since $H \notin \mathcal{M}_T$, the claim is not attainable by definition. This proves (1).

Because $\mathcal{M}_T \neq \mathbb{R}^n$, not every contingent claim is attainable. Hence the market is incomplete. This proves (2).

To prove (3), let $P_{\mathcal{M}_T} H$ denote the orthogonal projection of H onto $\mathcal{M}_T$, and define

$$y := H - P_{\mathcal{M}_T} H.$$

Then $y \in \mathcal{M}_T^{\perp}$. Since $H \notin \mathcal{M}_T$, we have $y \neq 0$. Moreover,

$$y \cdot H = y \cdot (P_{\mathcal{M}_T} H + y) = \|y\|^2 > 0,$$

because $P_{\mathcal{M}_T} H \in \mathcal{M}_T$ and $y \perp \mathcal{M}_T$.

Fix any $q^* \in \mathcal{Q}$, which exists by arbitrage-freeness. Since each component of q^* is strictly positive, there exists $\bar{\varepsilon} > 0$ such that

$$q_i^* \pm \varepsilon y_i > 0 \quad \text{for all } i$$

whenever $0 < \varepsilon < \bar{\varepsilon}$.

Because constant discounted claims are attainable, $\mathbf{1} \in \mathcal{M}_T$. Since $y \in \mathcal{M}_T^{\perp}$,

$$\sum_{i=1}^{n} y_i = y \cdot \mathbf{1} = 0.$$

Hence for $0 < \varepsilon < \bar{\varepsilon}$, the vectors

$$q^{\pm} := q^* \pm \varepsilon y$$

belong to Δ_{++}^{n-1}.

Now let $X \in \mathcal{M}_T$. Since $y \perp \mathcal{M}_T$,

$$E_{q^{\pm}}[X] = E_{q^*}[X] \pm \varepsilon\, y \cdot X = E_{q^*}[X] = \pi_0(X).$$

Therefore $q^{\pm} \in \mathcal{Q}$. Since $y \neq 0$, we have $q^+ \neq q^-$. Thus $\mathcal{Q}$ contains at least two distinct elements. This proves (3).

Finally,

$$E_{q^{\pm}}[H] = E_{q^*}[H] \pm \varepsilon\, y \cdot H = E_{q^*}[H] \pm \varepsilon \|y\|^2.$$

Hence

$$E_{q^+}[H] \neq E_{q^-}[H],$$

so $\Pi(H)$ contains at least two distinct prices.

It remains to show that $\Pi(H)$ is a compact interval. The set $\mathcal{Q}$ is the intersection of the simplex Δ_{++}^{n-1} with an affine subspace defined by the linear pricing restrictions

$$E_q[X] = \pi_0(X), \qquad X \in \mathcal{M}_T.$$

Equivalently, replacing Δ_{++}^{n-1} by its closure Δ_{+}^{n-1} does not change compactness, and $\mathcal{Q}$ is a convex subset of a compact set, hence bounded; the pricing restrictions define a closed affine

set. Thus the closure of $\mathcal{Q}$ is compact and convex, and because $q \mapsto E_q[H]$ is linear, its image is a compact interval in $\mathbb{R}$. Since $\Pi(H)$ contains at least two distinct values, this interval is nondegenerate. Therefore

$$\Pi(H) = [\underline{\pi}(H), \overline{\pi}(H)] \quad \text{with} \quad \underline{\pi}(H) < \overline{\pi}(H).$$

This proves (4). □

Non-perfect correlation with traded assets is neither necessary nor sufficient for non-attainability; the relevant condition is failure of spanning of the event payoff by the marketed subspace.

Remark 1 (Continuous-time analogue)**.** *In a standard semimartingale model satisfying NFLVR, the arbitrage-free prices of a bounded claim H are given by*

$$\{E_Q[H] : Q \in \mathcal{M}_e\},$$

where $\mathcal{M}_e$ is the set of equivalent local martingale measures. If H is unattainable, the market is incomplete and the arbitrage-free price is generally not unique. Proposition 1 states the same spanning logic in a finite-state discounted setting, where the proof is fully explicit.

Why this matters. Proposition 1 shows that the prediction-market event claim is priced in an incomplete market whenever its payoff is not spanned by traded claims. In that case, no-arbitrage does not pin down a unique pricing measure or a unique price; it delivers a *set* of admissible pricing measures and an interval of admissible prices. The interpretation problem identified by Manski (2006) therefore has a precise asset-pricing counterpart: the observed market price need not equal the physical event probability, because it is generated under some admissible pricing measure rather than directly under the physical measure. A tractable structural pricing model must therefore impose an additional *selection rule* within the no-arbitrage set. Section 3.3 proposes such a selection rule.

Remark 2 (Comparison to incomplete-market pricing approaches)**.** *Several frameworks exist for pricing in incomplete markets: good-deal bounds (Cochrane and Saa-Requejo, 2000), the minimal entropy martingale measure (Frittelli, 2000), and utility-based pricing. The Wang transform offers a complementary approach with a key advantage: it is a single-parameter model with an economically interpretable parameter (λ = pricing wedge), which facilitates cross-sectional estimation and comparative analysis. Its axiomatic foundations in the theory of distortion risk measures (Wang, Young, and Panjer, 1997; Yaari, 1987) provide rigorous justification.*

3.2 The Log-Odds State Variable

Before decomposing prices, we need a state variable in which standard stochastic tools apply. The contract price $p_t \in (0, 1)$ is bounded, violating the support assumptions of standard diffusion models. The resolution is the log-odds (logit) transformation:

$$x_t = \ln\left(\frac{p_t}{1-p_t}\right) = \text{logit}(p_t), \quad x_t \in \mathbb{R}$$

with inverse $p_t = \sigma(x_t) = (1 + e^{-x_t})^{-1}$.

This is the prediction market analog of the log-price transformation $y_t = \ln S_t$ in equity option pricing: just as $\ln S_t$ maps $(0, \infty) \to \mathbb{R}$ and yields geometric Brownian motion as the simplest tractable model, $\text{logit}(p_t)$ maps $(0, 1) \to \mathbb{R}$ and yields a "logistic diffusion" as the natural baseline.

Why log-odds is natural. The claim rests on two theoretical arguments. First, an *Itô's lemma argument*: if p_t follows a bounded diffusion with volatility $\sigma_p(p_t, t)$, the Jacobian $g'(p) = 1/(p(1-p))$ of the logit automatically compensates for boundary compression. If $\sigma_p(p) = \sigma \cdot p(1-p)$ (the martingale-consistent form), then x_t has constant diffusion coefficient σ—the simplest possible dynamics. The full Itô expansion of $x_t = \text{logit}(p_t)$ yields $dx_t = [\mu_p/(p(1-p)) + \frac{1}{2}\sigma^2(2p-1)]dt + \sigma\, dW_t$, confirming that the logit transformation produces a constant diffusion coefficient σ while introducing a state-dependent drift. The baseline pricing formula assumes the simplified case where the drift contribution is negligible relative to the diffusion term. Second, a *statistical argument*: the logit is the canonical link function for the Bernoulli exponential family in generalized linear models (McCullagh and Nelder, 1989); prediction market contracts *are* binary outcomes, so the logit is the information-theoretically natural parameterization. The empirical validation of these arguments is presented in Section 4.2.

Based on the empirical support, I posit a jump-diffusion in log-odds space, following the specification of Dalen (2025):

$$dx_t = \mu(x_t, t)\, dt + \sigma_b(x_t, t)\, dW_t + J_t\, dN_t$$

where σ_b is the *belief volatility*, J_t is the jump amplitude, and N_t is a Poisson process with intensity η. To derive the baseline pricing formula, I introduce a latent information state S_t that summarizes all available information about event E. Under the physical measure P, the terminal state follows $S_T \mid S_t \sim N(S_t, \sigma^2\tau)$ in the constant-volatility, no-jump case. The event occurs if and only if $S_T > 0$. The physical probability of the event is therefore:

$$p_t^* = P(S_T > 0 \mid S_t) = \Phi\left(\frac{S_t}{\sigma\sqrt{\tau}}\right)$$

This is the prediction market analog of the Black-Scholes derivation: just as the equity option formula arises from log-normal terminal stock prices, the binary contract formula arises from Gaussian terminal information states. In practice, S_t is not directly observed; the log-odds $x_t = \text{logit}(p_t)$ serves as the empirical proxy. The logit and probit functions are approximately proportional ($\text{logit}(p) \approx (\pi/\sqrt{3}) \cdot \Phi^{-1}(p)$), so the qualitative properties documented in Section 4 carry over.

3.3 The Wang Transform as a Maintained Selection Rule

Proposition 1 shows that if the event payoff $H = \mathbf{1}_E$ is not spanned by traded claims, then no-arbitrage leaves a *set* $\mathcal{Q}$ of admissible pricing measures rather than a unique one. The remaining problem is therefore a *selection problem*: which pricing measure within $\mathcal{Q}$ maps the physical event probability into the observed market price?

I do not claim that no-arbitrage or market completeness uniquely selects the Wang transform. Instead, I impose a one-factor Gaussian selection family within the incomplete-market set of admissible pricing measures. Under this maintained family, the Wang transform arises exactly.

Latent threshold representation. Let the event occur if and only if a latent terminal information index crosses a threshold:

$$E = \{Y_T \geq 0\}.$$

Assume that under the physical measure P,

$$Y_T = m_t + \Sigma_{t,T} Z_{t,T}, \qquad Z_{t,T} \mid \mathcal{F}_t \sim N(0,1),$$

where $m_t \in \mathbb{R}$ and $\Sigma_{t,T} > 0$ are $\mathcal{F}_t$-measurable.

Then the physical event probability is

$$p_t^* = P(E \mid \mathcal{F}_t) = P\left(Z_{t,T} \geq -\frac{m_t}{\Sigma_{t,T}} \,\middle|\, \mathcal{F}_t\right) = \Phi\left(\frac{m_t}{\Sigma_{t,T}}\right).$$

Proposition 2 (Exact Wang pricing under a Gaussian selection rule)**.** *Suppose the event admits the latent threshold representation above. Assume further that, within the incomplete-market set of admissible pricing measures, the market selects the one-parameter family* $\{Q^\lambda\}_{\lambda \in \mathbb{R}}$ *defined by*

$$\frac{dQ^\lambda}{dP}\Big|_{\mathcal{F}_T} = \exp\left(\lambda Z_{t,T} - \frac{1}{2}\lambda^2\right).$$

Then the observed market price of the binary event claim is

$$p_t^{\text{mkt}} = Q^\lambda(E \mid \mathcal{F}_t) = \Phi\left(\Phi^{-1}(p_t^*) + \lambda\right). \tag{1}$$

Proof. By construction,

$$p_t^* = P(E \mid \mathcal{F}_t) = P(Y_T \geq 0 \mid \mathcal{F}_t) = P\left(Z_{t,T} \geq -\frac{m_t}{\Sigma_{t,T}} \,\middle|\, \mathcal{F}_t\right) = \Phi\left(\frac{m_t}{\Sigma_{t,T}}\right).$$

Hence

$$\Phi^{-1}(p_t^*) = \frac{m_t}{\Sigma_{t,T}}.$$

Under the exponential tilt

$$\frac{dQ^\lambda}{dP}\Big|_{\mathcal{F}_T} = \exp\left(\lambda Z_{t,T} - \frac{1}{2}\lambda^2\right),$$

the standard normal factor is shifted from $N(0,1)$ to $N(\lambda,1)$. Therefore

$$p_t^{\mathrm{mkt}} = Q^\lambda(E \mid \mathcal{F}_t) = Q^\lambda\left(Z_{t,T} \geq -\frac{m_t}{\Sigma_{t,T}} \,\middle|\, \mathcal{F}_t\right) = \Phi\left(\frac{m_t}{\Sigma_{t,T}} + \lambda\right).$$

Substituting $\Phi^{-1}(p_t^*) = m_t/\Sigma_{t,T}$ yields

$$p_t^{\mathrm{mkt}} = \Phi\Big(\Phi^{-1}(p_t^*) + \lambda\Big). \qquad \square$$

Remark 3 (Generalized probit-linear family)**.** *The Wang transform is the variance-preserving special case of a broader Gaussian family. If the selected pricing measure changes both the conditional mean and variance of the latent index so that*

$$Y_T \mid \mathcal{F}_t \sim N(m_t + \delta_t, \tilde{\Sigma}_{t,T}^2),$$

then

$$p_t^{\mathrm{mkt}} = \Phi\left(\frac{m_t + \delta_t}{\tilde{\Sigma}_{t,T}}\right) = \Phi\Big(\alpha_t + \beta_t \Phi^{-1}(p_t^*)\Big),$$

where

$$\alpha_t := \frac{\delta_t}{\tilde{\Sigma}_{t,T}}, \qquad \beta_t := \frac{\Sigma_{t,T}}{\tilde{\Sigma}_{t,T}}.$$

The Wang transform is the exact restriction $\beta_t = 1$, i.e. the selected pricing measure shifts the latent mean but preserves conditional variance.

Remark 4 (When jumps break exact Wang)**.** *If the latent terminal index contains an additional jump component,*

$$Y_T = m_t + \Sigma_{t,T} Z_{t,T} + J_{t,T},$$

then under the same Gaussian tilt of $Z_{t,T}$,

$$p_t^{\mathrm{mkt}} = E\left[\Phi\left(\frac{m_t + J_{t,T}}{\Sigma_{t,T}} + \lambda\right) \middle| \mathcal{F}_t\right].$$

This is generally a mixture of Wang curves rather than a single Wang transform of p_t^*. Exact Wang therefore requires the conditional Gaussian latent-index specification or another restriction that preserves the probit-linear family.

Axiomatic background. The Wang transform is not chosen arbitrarily. It is a standard distortion operator in the theory of distortion risk measures (Wang, Young, and Panjer, 1997; Yaari, 1987; Denneberg, 1994) and can be justified from multiple directions: as a probability distortion satisfying the usual monotonicity and boundary conditions, as the variance-preserving member of the Gaussian probit-linear family, and as the survival-function representation induced by an exponential tilt of a latent normal factor. For present purposes, the key point is narrower: in an incomplete market, Wang is not forced by no-arbitrage, but it is a tractable, economically interpretable, and exactly derived selection rule under the maintained Gaussian specification. A detailed axiomatic treatment is provided in Appendix A.

Measure-change interpretation. In the maintained one-factor Gaussian setting above, the selected pricing family is generated by an exponential tilt of the latent standardized event factor:

$$\frac{dQ^\lambda}{dP}\Big|_{\mathcal{F}_T} = \exp\Big(\lambda Z_{t,T} - \frac{1}{2}\lambda^2\Big).$$

This is the single-factor Gaussian analogue of a Girsanov shift: it changes the conditional mean of the latent event factor while preserving its variance. The Wang transform is therefore exact *within this maintained family.* I do not claim that arbitrary continuous-time incomplete-market models imply Wang, nor that the Wang transform is the unique arbitrage-free pricing rule outside this specification. Pelsser (2008) showed that the Wang transform is not consistent with arbitrage-free pricing for general stochastic processes; that critique targets dynamic replication in complete markets and does not apply to the static, incomplete-market setting of this paper.

Consistency with financial theory. Kijima and Muromachi (2008) showed that the Wang transform can be derived from Bühlmann's economic premium principle, establishing an equilibrium foundation for risks correlated with the aggregate economy. The parameter λ is economically interpretable: under the Bühlmann benchmark, it equals the product of aggregate risk aversion and the standard deviation of aggregate risk (Section 3.6). For prediction market events that are largely uncorrelated with aggregate risk, the aggregate-risk component of λ may be small. In that case, the estimated selection parameter should be interpreted as a composite object within the maintained pricing family, potentially reflecting inventory risk, adverse selection, capital constraints, ambiguity, or other market-specific frictions (Sections 5.3–5.4).

Corollary 1 (Pricing wedge properties)**.** *Under*

$$p^{\text{mkt}} = \Phi(\Phi^{-1}(p^*) + \lambda), \qquad \lambda > 0,$$

the following hold:

1. $p^{\text{mkt}} > p^*$ *for all* $p^* \in (0, 1)$.
2. *The level overpricing* $\Delta(p^*) = p^{\text{mkt}} - p^*$ *is maximized at*

$$p^* = \Phi(-\lambda/2).$$

3. *The ratio* $R(p^*) = p^{\text{mkt}}/p^*$ *is strictly decreasing in* p^*.

This overpricing-ratio pattern is qualitatively consistent with the ***favorite-longshot bias*** *documented by Bürgi, Deng, and Whelan (2025) across 300,000+ Kalshi contracts and by Snowberg and Wolfers (2010) across 6.4 million horse racing starts.*

Proof. Let $z = \Phi^{-1}(p^*)$. Then $p^{\text{mkt}} = \Phi(z + \lambda)$. Part (1) follows immediately from the strict monotonicity of Φ.

For part (2), define

$$\Delta(z) = \Phi(z + \lambda) - \Phi(z).$$

Then

$$\Delta'(z) = \phi(z+\lambda) - \phi(z).$$

Setting $\Delta'(z) = 0$ yields

$$\phi(z+\lambda) = \phi(z) \quad \Longleftrightarrow \quad (z+\lambda)^2 = z^2 \quad \Longleftrightarrow \quad z = -\lambda/2.$$

Hence the maximizer is $p^* = \Phi(-\lambda/2)$.

For part (3), define

$$R(z) = \frac{\Phi(z+\lambda)}{\Phi(z)}.$$

Then

$$R'(z) = \frac{\phi(z+\lambda)\Phi(z) - \Phi(z+\lambda)\phi(z)}{\Phi(z)^2}.$$

Since the inverse Mills ratio $h(z) := \phi(z)/\Phi(z)$ is strictly decreasing in z and $z+\lambda > z$, we have

$$\frac{\phi(z+\lambda)}{\Phi(z+\lambda)} < \frac{\phi(z)}{\Phi(z)},$$

which implies $R'(z) < 0$. Since $p^* = \Phi(z)$ is increasing in z, $R(p^*)$ is strictly decreasing in p^*. □

The monotonicity of the overpricing ratio is a known property of concave probability distortion functions (Yaari, 1987); the contribution here is that under the maintained Gaussian selection rule, this pattern emerges with a single interpretable parameter λ.

Remark 5 (Relation to alternative FLB explanations)**.** *The FLB has been attributed to risk-loving preferences (Ali, 1977), probability misperceptions via prospect theory (Snowberg and Wolfers, 2010), adverse selection among informed bettors (Shin, 1991, 1993), and noise-to-signal ratios (Ottaviani and Sorensen, 2010). The Wang transform explanation is complementary: it provides a structural, risk-premium-based mechanism through the supply side. Whelan's (2024) competitive bookmaker model, in which FLB arises from risk-averse market makers, is the closest analog—λ in the Wang transform can be interpreted as the aggregate risk aversion of market makers.*

Remark 6 (Complement Invariance, Unified Order Books, and Contract Architecture)**.** *The Wang transform specification in equation (1) is not framing-invariant: relabeling event E as $\neg E$ transforms λ to $-\lambda$. This is a known property of directional distortion operators (Denneberg, 1994). However, the framing concern presupposes that YES and NO are economically distinct positions requiring separate pricing—a premise that prediction market architecture directly contradicts.*

Both Kalshi and Polymarket implement unified order books: buying YES at price p and selling NO at price $1-p$ are computationally identical operations executed against the same book. On Kalshi, a YES buyer paying \$c is matched with a NO buyer paying \$(1-c), and the combined \$1 is held by the clearinghouse until settlement. On Polymarket, the `splitPosition()` *smart contract*

mints exactly one YES token and one NO token per 1 USDC deposited; `mergePositions()` *reverses the operation. These mechanisms ensure* $p_{YES} + p_{NO} \equiv 1$ *not as an equilibrium condition or an arbitrage result, but as a mathematical property of the contract creation mechanism—the identity cannot be violated even instantaneously.*

This architecture means the observed market price is a single scalar $p \in (0, 1)$ *without inherent directional content: the label "YES" is a naming convention, not an economic primitive. The framing non-invariance of the Wang transform is therefore primarily an artifact of parameterization choice rather than a fundamental economic limitation, though residual framing effects—arising from trader cognition, interface design, or liquidity asymmetries between YES and NO order books—cannot be fully ruled out on theoretical grounds. The complement-invariant specification in Section 5.6 provides an empirical assessment of the magnitude of such effects.*

Note that executable quotes (best ask for YES plus best ask for NO) can exceed 1 due to the bid-ask spread. The spread compensates market makers for immediacy provision—a liquidity premium that is economically and empirically distinct from the mid-price probability distortion captured by λ*.*

3.4 Testable Predictions

The framework generates two testable predictions and a set of quantitative benchmarks.

Prediction 1: Favorite-longshot bias. If $\lambda > 0$, the Corollary predicts that market prices systematically overstate event probabilities, with the overpricing ratio decreasing in p^*. This is testable by comparing market prices to realized resolution frequencies.

Prediction 2: Cross-platform spread. If two platforms have different pricing wedge loadings ($\lambda_K \neq \lambda_P$), then systematic price differences arise: $p_K - p_P = \Phi(\Phi^{-1}(p^*) + \lambda_K) - \Phi(\Phi^{-1}(p^*) + \lambda_P)$. This is testable using aligned Polymarket-Kalshi data.

Numerical benchmarks. To build intuition, Table 1 shows the Wang transform for several values of λ.

True prob. p^*	$\lambda = 0$	$\lambda = 0.1$	$\lambda = 0.3$	$\lambda = 0.5$
0.05	0.050	0.061	0.089	0.126
0.10	0.100	0.119	0.163	0.217
0.20	0.200	0.229	0.294	0.366
0.30	0.300	0.336	0.411	0.490
0.50	0.500	0.540	0.618	0.691
0.70	0.700	0.734	0.795	0.847
0.90	0.900	0.916	0.943	0.963
0.95	0.950	0.959	0.974	0.984

Table 1: Wang-distorted probabilities $\tilde{p} = \Phi(\Phi^{-1}(p^*) + \lambda)$. The distortion is strongest at low probabilities and weakest at high probabilities, generating the FLB pattern.

3.5 The Complete Pricing Model

The framework has three conceptually distinct layers: an observed quote representation, a latent physical event model, and a maintained pricing-measure selection rule.

Definition 1 (Prediction market pricing model)**.** *Let $p_t^{\text{mkt}} \in (0,1)$ denote the observed market price of a binary contract on event E with time-to-expiry $\tau = T - t$.*

Layer 1 — Observation space. *Observed quotes live on the bounded interval $(0,1)$. For empirical time-series analysis, I map them into log-odds space:*

$$x_t^{\text{mkt}} = \text{logit}(p_t^{\text{mkt}}) = \ln\left(\frac{p_t^{\text{mkt}}}{1 - p_t^{\text{mkt}}}\right).$$

This transformation is useful for descriptive dynamics and empirical stylized facts, but it is not the primitive object from which the Wang pricing equation is derived.

Layer 2 — Physical event model. *The event is generated by a latent threshold factor:*

$$E = \{Y_T \geq 0\}, \qquad Y_T = m_t + \Sigma_{t,T} Z_{t,T}, \qquad Z_{t,T} \mid \mathcal{F}_t \sim N(0,1).$$

Hence the physical event probability is

$$p_t^* = P(E \mid \mathcal{F}_t) = \Phi\left(\frac{m_t}{\Sigma_{t,T}}\right).$$

Layer 3 — Pricing-measure selection rule. *Because the event claim is unattainable, no-arbitrage leaves a non-singleton set of admissible pricing measures. I impose the maintained one-parameter family*

$$\frac{dQ^\lambda}{dP}\Big|_{\mathcal{F}_T} = \exp\left(\lambda Z_{t,T} - \frac{1}{2}\lambda^2\right),$$

which yields the observed market price

$$p_t^{\text{mkt}} = Q^\lambda(E \mid \mathcal{F}_t) = \Phi\Big(\Phi^{-1}(p_t^*) + \lambda\Big).$$

Thus uniqueness holds only conditional on the maintained family*: the Wang transform is not the unique arbitrage-free price system, but the unique variance-preserving member of the Gaussian probit-linear selection family.*

Forward pricing. Given a physical probability estimate $\hat{p}_t^*$ and a calibrated pricing-wedge parameter $\hat{\lambda}$,

$$\hat{p}_t^{\text{mkt}} = \Phi\Big(\Phi^{-1}(\hat{p}_t^*) + \hat{\lambda}\Big).$$

Inverse mapping. Given an observed market price p_t^{mkt} and a calibrated $\hat{\lambda}$,

$$\hat{p}_t^* = \Phi\Big(\Phi^{-1}(p_t^{\text{mkt}}) - \hat{\lambda}\Big).$$

Calibration. The parameter λ is estimated from a cross-section of resolved contracts (p_i^{mkt}, y_i), where $y_i \in \{0, 1\}$ is the resolution outcome:

$$\hat{\lambda}_{\text{MLE}} = \arg\max_{\lambda} \sum_{i=1}^{N} \Big[y_i \ln \Phi\Big(\Phi^{-1}(p_i^{\text{mkt}}) - \lambda\Big) + (1 - y_i) \ln\Big(1 - \Phi\Big(\Phi^{-1}(p_i^{\text{mkt}}) - \lambda\Big)\Big)\Big].$$

Standard errors are computed from the observed Fisher information (numerical Hessian). The likelihood ratio test against $H_0 : \lambda = 0$ (prices are perfectly calibrated probabilities) has one degree of freedom.

Model sensitivities. The Wang transform implies analytic sensitivities of the market price to model inputs:

- *Lambda sensitivity*: $\dfrac{\partial p^{\text{mkt}}}{\partial \lambda} = \phi\Big(\Phi^{-1}(p^*) + \lambda\Big) > 0$. The market price is monotonically increasing in the pricing wedge, with maximum sensitivity at $p^* \approx 0.5$.
- *Probability sensitivity*: $\dfrac{\partial p^{\text{mkt}}}{\partial p^*} = \dfrac{\phi(\Phi^{-1}(p^*) + \lambda)}{\phi(\Phi^{-1}(p^*))}$, the local amplification factor of the Wang distortion.
- *Overpricing*: $\Delta p = p^{\text{mkt}} - p^*$, maximized near $p^* = 0.5$ (Corollary 1).

Summary. The complete model reduces prediction market pricing to three objects: an observation-space representation (log-odds of quoted prices), a latent physical event model (Gaussian threshold factor), and a maintained pricing-measure selection rule (one-parameter exponential tilt yielding the Wang transform). It takes a market price as input and outputs a risk-adjusted probability; or takes a probability forecast and outputs a model-implied fair price. The empirical calibration of λ is the subject of Section 5.

3.6 Economic Interpretation and Scope of λ

Within the model of Sections 3.3–3.5, λ is the selection parameter of the maintained pricing family $\{Q^\lambda\}$. It measures the shift, in probit space, between the physical event probability and the market-selected pricing measure:

$$p_t^{\text{mkt}} = \Phi\Big(\Phi^{-1}(p_t^*) + \lambda_t\Big).$$

Equivalently, λ_t is the mean shift applied to the standardized latent event factor under the selected pricing measure. In this sense, λ is a structural object *within the maintained Gaussian selection family*, not merely an arbitrary calibration constant.

A benchmark equilibrium mapping. To connect λ to standard asset-pricing objects, suppose the latent event index admits the decomposition

$$Y_T = m_t + \beta_t F_{t,T} + \varepsilon_{t,T},$$

where, conditional on $\mathcal{F}_t$,

$$F_{t,T} \sim N(0,1), \qquad \varepsilon_{t,T} \sim N(0, \sigma^2_{\varepsilon,t}),$$

and $F_{t,T}$ and $\varepsilon_{t,T}$ are independent. Here $F_{t,T}$ is a priced aggregate factor and $\varepsilon_{t,T}$ is idiosyncratic event risk. Then

$$\Sigma^2_{t,T} = \beta_t^2 + \sigma^2_{\varepsilon,t}, \qquad p_t^* = \Phi\left(\frac{m_t}{\Sigma_{t,T}}\right).$$

Now suppose the pricing kernel tilts the priced factor $F_{t,T}$ according to

$$\frac{dQ^\theta}{dP}\Big|_{\mathcal{F}_T} = \exp\left(-\theta_t F_{t,T} - \frac{1}{2}\theta_t^2\right).$$

Then under Q^θ,

$$F_{t,T} \mid \mathcal{F}_t \sim N(-\theta_t, 1),$$

so the latent event index satisfies

$$Y_T \mid \mathcal{F}_t \sim N(m_t - \beta_t\theta_t, \Sigma^2_{t,T}).$$

Therefore

$$p_t^{\text{mkt}} = Q^\theta(E \mid \mathcal{F}_t) = \Phi\left(\frac{m_t - \beta_t\theta_t}{\Sigma_{t,T}}\right) = \Phi\left(\Phi^{-1}(p_t^*) + \lambda_t\right),$$

with

$$\lambda_t = -\frac{\beta_t\theta_t}{\Sigma_{t,T}}. \tag{2}$$

This benchmark gives λ_t a standard economic interpretation: it is proportional to the event's loading on a priced factor (β_t) and the market price of that factor (θ_t), scaled by total event uncertainty ($\Sigma_{t,T}$). The sign of λ_t depends on whether the event payoff loads positively or negatively on states with high marginal value: $\lambda_t > 0$ when the event tends to pay in high-marginal-utility states; $\lambda_t < 0$ when it pays more in good states.

Why the benchmark may predict small aggregate-risk loadings. For many prediction-market events—geopolitical outcomes, sporting results, scientific milestones—exposure to broad aggregate risk may be limited, implying a small β_t and hence a small aggregate-risk component of λ_t. Persistent positive estimates of $\hat{\lambda}$ in the data therefore suggest that additional structural channels contribute to the market's selection rule, including

market-maker inventory risk (Whelan, 2024; Avellaneda and Stoikov, 2008), adverse selection (Shin, 1991), intermediary capital constraints (He and Krishnamurthy, 2013), capital lockup opportunity costs (Page and Clemen, 2013), and ambiguity aversion (Gilboa and Schmeidler, 1989). The role of the Wang specification is not to claim that these mechanisms are identical, but to provide a common one-parameter selection rule that captures the resulting probability compression in a tractable way.

Empirical signatures across structural channels. The empirical patterns documented in Sections 5–6 are consistent with multiple structural sources of λ and help distinguish among them:

Time-varying decay. The pricing wedge declines from $\hat{\lambda}(0.05) = 0.173$ at market opening to $\hat{\lambda}(0.80) = 0.037$ near resolution (Section 5.4). Under equation (2), this is predicted whenever residual event uncertainty $\Sigma_{t,T}$ decreases as information arrives, reducing $|\lambda_t|$ regardless of its structural source.

Volume effect. The negative coefficient on ln(Volume) in the hierarchical model (Table 5) is predicted by market-structure models—inventory risk, adverse selection, capital constraints—but is *not* predicted by pure cognitive bias explanations, which imply λ should be invariant to liquidity.

Category heterogeneity. Sports and politics markets ($\hat{\lambda} \approx 0.05$–$0.07$, insignificant) show near-zero pricing wedges, while thinner categories (crypto, science/tech) exhibit large significant wedges ($\hat{\lambda} \approx 0.25$–$0.28$). The pricing wedge is competed away in liquid, well-followed markets but persists where risk-bearing capacity is scarce.

Play-money sign reversal. Manifold ($\hat{\lambda} = -0.22$) vs. all real-money platforms ($\hat{\lambda} > 0$) provides a natural control: removing financial stakes eliminates the risk-bearing cost channel while preserving cognitive biases, consistent with the sign flip.

Scope and identification. The structural model clarifies what λ means within the maintained Gaussian family. However, price–outcome data alone do not separately identify the structural source of the estimated $\hat{\lambda}$. In particular, the empirical calibration in Sections 5–6 recovers the composite selection parameter consistent with the maintained Wang specification, but does not by itself decompose $\hat{\lambda}$ into aggregate-risk, inventory, adverse-selection, ambiguity, or microstructure components. That decomposition requires additional data or exogenous variation (Section 8).

4 Data and Descriptive Statistics

4.1 Data Sources and Sample Construction

Primary sample (Polymarket). I collect contract-level data from Polymarket via its public Gamma API (market metadata) and CLOB API (hourly price histories). The Gamma API provides market-level attributes including resolution outcome, trading volume, creation and resolution timestamps, and contract descriptions. The CLOB API provides hourly mid-price snapshots for each contract's trading history.

The raw sample consists of 15,525 resolved binary markets from February 24 to March 24, 2026, spanning 28 days. I apply the following filters: (i) exclude pure randomness markets (e.g., crypto "up or down" on 5-minute intervals, esports over/under markets); (ii) require at least 4 hourly price observations to ensure meaningful price dynamics; and (iii) restrict to $p \in (0.02, 0.98)$ to avoid boundary artifacts where the logit transformation amplifies noise. The final Polymarket estimation sample contains $N = 13{,}738$ contracts spanning sports, politics, crypto, economics, and other categories. For each contract, I extract the *opening price* (first 5% of the market's lifetime) and the *resolution outcome* (1 if the event occurred, 0 otherwise). For the time-varying analysis (Section 5.4), I also extract prices at ten percentage-of-lifetime horizons and at fixed clock-time intervals before resolution.

For the cross-platform analysis (Section 5.5), I use a separate dataset: aligned daily prices for the Greenland acquisition contract on both Polymarket and Kalshi, collected via their respective APIs.

Extended sample (cross-platform). For the cross-platform validation (Section 6), I assemble four additional data sources: (i) Kalshi settled binary markets ($N = 271{,}699$) from both the historical and live API tiers, spanning 2021–2026, filtered to volume ≥ 100 contracts and $p \in (0.02, 0.98)$; (ii) a Polymarket 2025 sample ($N = 985$) from HuggingFace (LightningRodLabs/outcome-rl-test-dataset), covering February–March 2025; (iii) multi-platform forecasting data ($N = 4{,}887$) from HuggingFace (YuehHanChen/forecasting), covering Metaculus, Good Judgment Open, INFER (formerly CSET-foretell), Polymarket, and Manifold questions from 2015–2024; and (iv) the primary Polymarket CLOB sample described above. The pooled cross-platform sample totals $N = 291{,}309$ resolved contracts.

4.2 Stylized Facts in Log-Odds Space

Before proceeding to the Wang transform calibration, I validate the log-odds state variable empirically. If log-odds is the correct state variable, its empirical properties should mirror those of equity log-returns, which have been extensively documented (Cont, 2001).

Table 2 compares distributional properties of raw price increments Δp_t and log-odds increments Δx_t for selected contracts.

Contract	N	Excess Kurtosis		Skewness	
		Δp	Δx	Δp	Δx
Fed No Change (hourly)	628	7.0	6.0	-0.50	-0.14
Fed Cut 25bps (hourly)	613	11.3	6.5	-0.24	-0.26
Greenland Acquisition (daily)	89	16.8	12.1	-2.02	-1.04

Table 2: Distributional properties of price increments. Log-odds increments have systematically lower excess kurtosis and reduced skewness, moving closer to the Gaussian benchmark.

Table 3 confirms the broader correspondence with equity log-return stylized facts.

Stylized Fact	Equity Log-Returns	Prediction Market Δx_t
Random walk	Log-prices: unit root (ADF); returns stationary	Levels x_t: unit root (ADF $t = -0.03$ to -2.23); Δx_t stationary
Fat tails	Excess kurtosis 3–50	Excess kurtosis 6–12
Vol. clustering	$\rho(\Delta r_t^2, \Delta r_{t-1}^2) > 0$	$\rho(\Delta x_t^2, \Delta x_{t-1}^2) = 0.13$–$0.14$***
Negative lag-1 AC	Bid-ask bounce	$\rho_1 = -0.11$ to -0.17***
Asymmetric vol.	Leverage effect	Event-driven spikes

Table 3: Stylized facts comparison. *** denotes significance at 1%. Log-odds increments exhibit the same statistical signature as equity log-returns.

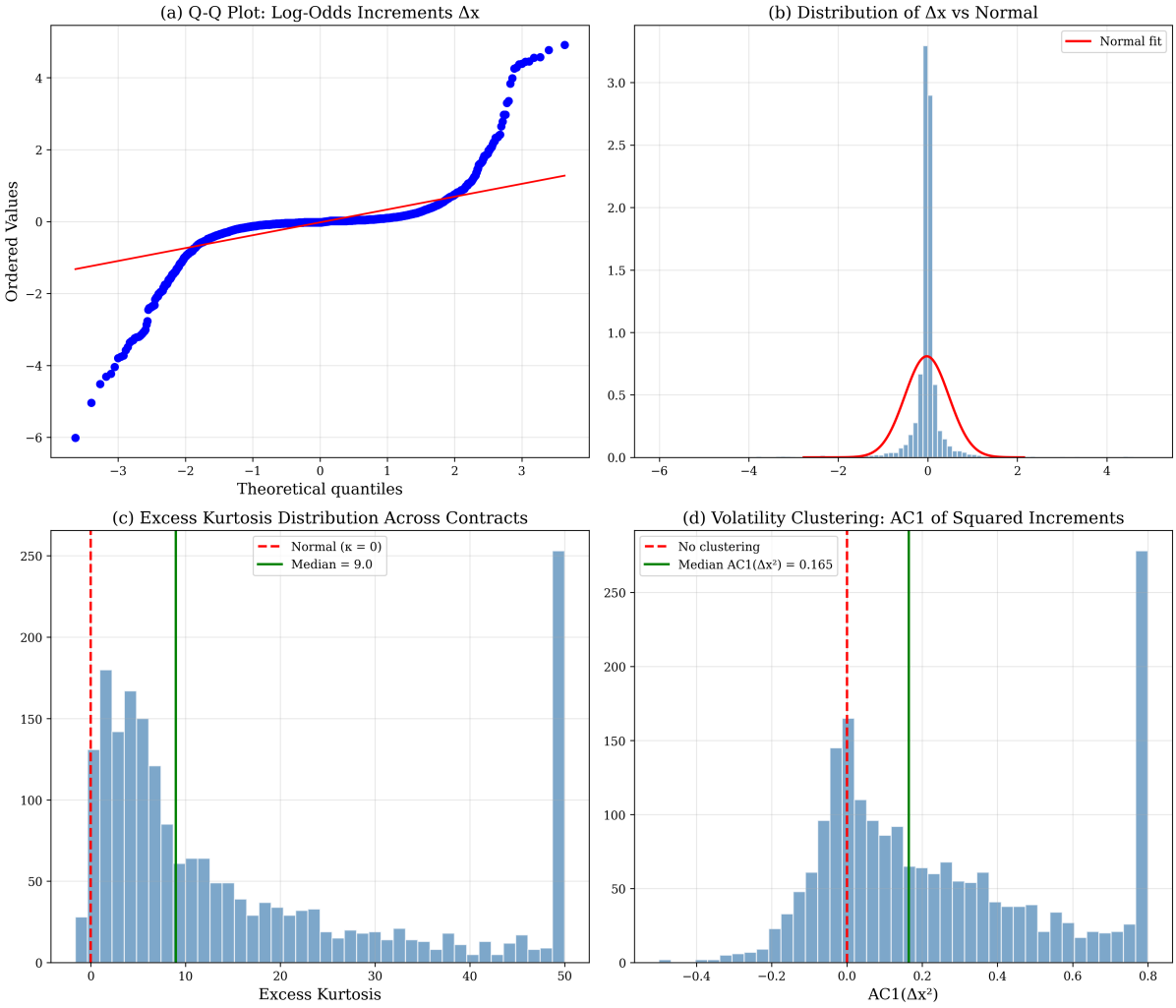

Figure 1: Log-odds stylized facts across 2,036 contracts (148,114 pooled increments). (a) Q-Q plot of Δx against the normal distribution. (b) Distribution of Δx with normal overlay. (c) Distribution of per-contract excess kurtosis. (d) Distribution of autocorrelation of squared increments (volatility clustering measure).

Pooled sample statistics. Across 2,036 contracts with sufficient price history (148,114 total log-odds increments), the median per-contract excess kurtosis is 9.0 (range 6–12 for well-traded contracts), confirming the fat-tail stylized fact. The median autocorrelation of squared increments—a nonparametric volatility clustering measure—is $\mathrm{AC}_1(\Delta x_t^2) = 0.16$, confirming significant ARCH effects. An Engle ARCH(1) LM test on the pooled sample is overwhelmingly significant ($p < 10^{-20}$). Ljung-Box tests reject white noise in both Δx ($Q_{10} = 91.3$, $p < 10^{-14}$) and Δx^2 ($Q_{10} = 131.0$, $p < 10^{-22}$).

These stylized facts—random-walk behavior in levels, fat tails, volatility clustering, and improved normality under the logit transformation—validate the log-odds state variable and support the diffusion-based modeling approach adopted in Section 3.2.

5 Empirical Results

5.1 Identification Strategy

The empirical objective is to estimate the *composite pricing-measure selection parameter* implied by the maintained Wang family, not to nonparametrically recover contract-level physical probabilities or to separately identify the structural source of λ from price–outcome data alone.

At the individual-contract level, the physical probability p_i^* is unobserved. Upon resolution one observes only the realized outcome $Y_i \in \{0, 1\}$, not the ex ante physical probability. Under the maintained Wang specification,

$$p_i^{\text{mkt}} = \Phi\Big(\Phi^{-1}(p_i^*) + \lambda_i\Big),$$

so equivalently

$$p_i^*(\lambda_i) = \Phi\Big(\Phi^{-1}(p_i^{\text{mkt}}) - \lambda_i\Big).$$

If $\lambda_i = \lambda$ is constant across contracts, the model implies

$$\Pr(Y_i = 1 \mid p_i^{\text{mkt}}) = \Phi\Big(\Phi^{-1}(p_i^{\text{mkt}}) - \lambda\Big),$$

which yields a Bernoulli likelihood for λ. The maximum-likelihood estimator therefore recovers the constant selection parameter that best rationalizes the cross-section of observed prices and realized outcomes under the maintained Wang family:

$$\hat{\lambda}_{\text{MLE}} = \arg\max_{\lambda} \sum_{i=1}^{N} \Big[y_i \ln \Phi\Big(\Phi^{-1}(p_i^{\text{mkt}}) - \lambda\Big) + (1 - y_i) \ln\Big(1 - \Phi\Big(\Phi^{-1}(p_i^{\text{mkt}}) - \lambda\Big)\Big)\Big].$$

Standard errors are computed from the observed Fisher information (numerical Hessian). A likelihood ratio test against $H_0 : \lambda = 0$ (prices are perfectly calibrated probabilities) has one degree of freedom.

This interpretation is important. Resolved-contract data identify a calibration relationship of the form $E[Y \mid p^{\text{mkt}}]$ under exchangeability-type assumptions. The Wang model maps that empirical relationship into an estimated λ, but the result remains model-based: the data do not, by themselves, decompose $\hat{\lambda}$ into aggregate-risk, inventory, adverse-selection, ambiguity, or microstructure components.

Binned calibration. As a descriptive nonparametric complement, I group contracts into quantile bins by market price, compute the empirical resolution rate $\hat{p}_k^*$ in each bin, and estimate $\hat{\lambda}_k = \Phi^{-1}(\bar{p}_k^{\text{mkt}}) - \Phi^{-1}(\hat{p}_k^*)$ with bootstrap standard errors (2,000 replications). This binned estimator is used as a calibration diagnostic and robustness check, not as the paper's primary structural identification device.

Table 4: Wang transform calibration: Polymarket contract-level data (March 13–24, 2026; $N = 2{,}460$ contracts with $p \in (0.05, 0.95)$). $\hat{\lambda}$ per bin via $\hat{\lambda} = \Phi^{-1}(\bar{p}^{\text{mkt}}) - \Phi^{-1}(\hat{p}^*)$. Bootstrap standard errors from 2,000 replications; MLE via Bernoulli log-likelihood with observed Fisher information.

$\bar{p}^{\text{mkt}}$	$\hat{p}^*$	N	$\hat{\lambda}$	SE	95% CI
0.079	0.061	164	0.138	0.159	[−0.174, 0.449]
0.151	0.110	164	0.194	0.136	[−0.073, 0.461]
0.217	0.161	174	0.207	0.116	[−0.020, 0.434]
0.262	0.195	154	0.224	0.115	[−0.001, 0.449]
0.331	0.345	171	−0.040	0.101	[−0.238, 0.158]
0.414	0.343	172	0.188	0.100	[−0.007, 0.383]
0.463	0.311	151	0.400	0.109	[0.186, 0.613]
0.489	0.303	251	0.489	0.084	[0.325, 0.654]
0.499	0.382	374	0.299	0.066	[0.169, 0.429]
0.505	0.514	142	−0.023	0.105	[−0.228, 0.183]
0.524	0.509	163	0.038	0.097	[−0.152, 0.228]
0.595	0.606	142	−0.026	0.108	[−0.238, 0.185]
0.771	0.822	163	−0.182	0.116	[−0.410, 0.045]
Binned (wtd)		**2,460**	**0.173**	0.029	[0.116, 0.230]
MLE		**2,460**	**0.176**	0.027	[0.123, 0.230]

Key findings:

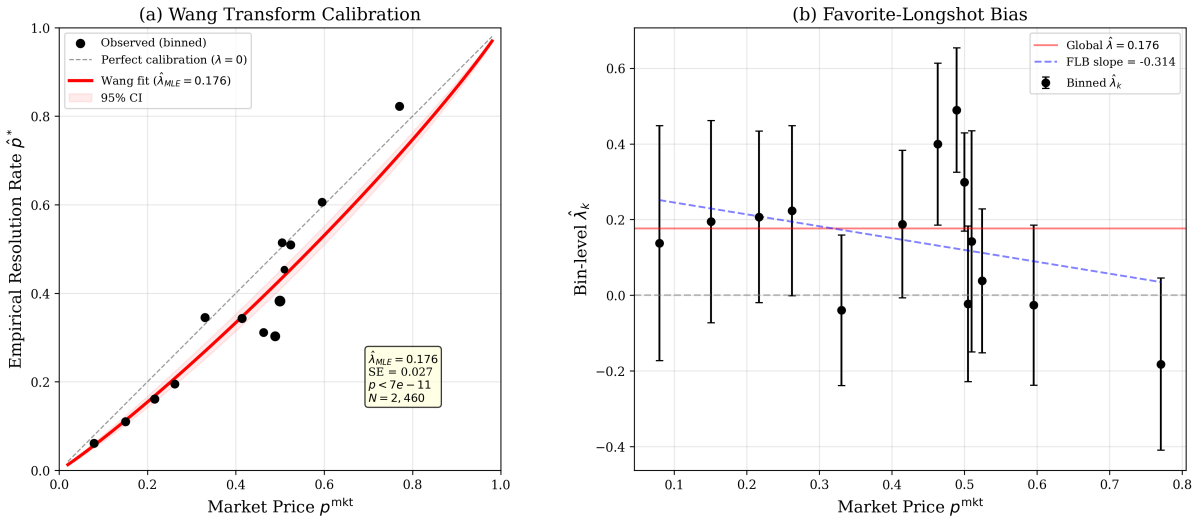

Figure 2: Wang transform calibration. (a) Market price vs. empirical resolution rate with MLE Wang fit ($\hat{\lambda} = 0.176$) and 95% CI. Points are quantile bins, sized by $\sqrt{N}$. The Wang curve (red) captures the systematic overpricing relative to perfect calibration (dashed). (b) Bin-level $\hat{\lambda}_k$ with 95% CIs, showing the FLB pattern (negative slope).

1. $\hat{\lambda}_{MLE} = 0.176$ **(SE** $= 0.027$**,** $p = 7.1 \times 10^{-11}$**).** The likelihood ratio test decisively rejects perfect calibration ($\chi^2 = 42.5$, $p < 10^{-10}$): prediction market prices embed a statistically significant positive pricing wedge.

2. **Consistent estimation.** The binned weighted average ($\hat{\lambda} = 0.173$, SE $= 0.029$) is within one standard error of the MLE, confirming robustness to the parametric functional form.

3. **Order of magnitude.** $\hat{\lambda} \approx 0.18$ is substantially below the catastrophe bond estimate of $\lambda \approx 0.45$ (Wang, 2004), consistent with prediction markets carrying lower pricing wedge than catastrophe insurance. This makes economic sense: much of event risk in prediction markets is idiosyncratic—unlike catastrophe exposure, which involves heavily correlated tail risk—though systematic factors (e.g., macro shocks, correlated political events) may introduce common risk components.

4. **Economic magnitude.** Under the estimated Wang transform, a contract with physical probability $p^* = 10\%$ trades at $\Phi(\Phi^{-1}(0.10) + 0.176) \approx 13.5\%$—an overpricing of 35%. At $p^* = 50\%$, the market price is $\Phi(0.176) \approx 57.0\%$—a 7.0 cent premium on a \$1 contract.

5. **Bootstrap confirmation.** A nonparametric bootstrap (2,000 replications) yields SE $= 0.028$ and a 95% CI of $[0.124, 0.232]$, virtually identical to the analytical estimates, confirming that the MLE is well-behaved and the Gaussian approximation is adequate.

6. **Expanded sample confirmation.** Extending the Polymarket sample from 12 to 28 days (February 24 to March 24, 2026; $N = 13{,}738$) yields $\hat{\lambda}_{MLE} = 0.166$ (SE $= 0.011$, LR $= 212.1$, $p < 10^{-15}$), confirming the primary estimate. The Wang correction reduces the Expected Calibration Error by 53.9% and the Brier score by 1.9% on this expanded sample.

7. **Non-constant $\hat{\lambda}$ across bins.** The bin-level estimates range from -0.18 to $+0.49$, with systematically higher values at lower prices—the favorite-longshot bias. This motivates the extended model in Section 5.3.

5.2 Cross-Sectional Determinants of λ

The non-constant $\hat{\lambda}$ across bins motivates allowing the pricing wedge to depend on contract characteristics. I estimate a one-stage probit model with known offset $z_i = \Phi^{-1}(p_i^{\text{mkt}})$, where the pricing wedge $\lambda_i = X_i\beta$ depends on contract characteristics:

$$\Pr(y_i = 1 \mid p_i^{\text{mkt}}, X_i) = \Phi\big(\Phi^{-1}(p_i^{\text{mkt}}) - X_i\beta\big)$$

This eliminates the generated-regressor problem inherent in the two-step approach (first estimating $\hat{\lambda}_i$ from smoothed resolution rates, then regressing on covariates), which understates standard errors by ignoring first-stage estimation error. The one-stage MLE jointly estimates

all parameters by maximum likelihood, yielding correct standard errors without the need for bootstrap correction.

Table 5: One-stage hierarchical Wang MLE. Dependent variable: binary outcome y_i. Model: $\Pr(y_i = 1) = \Phi(\Phi^{-1}(p_i^{\text{mkt}}) - X_i\beta)$. Fisher and sandwich robust standard errors reported.

	Full sample ($N = 13{,}274$)			12-day subsample ($N = 2{,}134$)		
Variable	Coef.	SE (Fisher)	SE (Robust)	Coef.	SE (Fisher)	SE (Robust)
Constant	0.259^{***}	(0.063)	(0.064)	0.387^{**}	(0.144)	(0.146)
ln(Volume)	-0.072^{***}	(0.005)	(0.005)	-0.100^{***}	(0.013)	(0.013)
ln(Duration)	0.143^{***}	(0.012)	(0.012)	0.176^{***}	(0.024)	(0.024)
$\lvert p - 0.5\rvert$ (extremity)	-0.477^{***}	(0.100)	(0.099)	-0.698^{**}	(0.252)	(0.259)
Spread	0.127	(0.384)	(0.423)	-10.000^{*}	(4.466)	(5.893)
Log-likelihood			−7,937.9			−1,263.5
LR test ($\lambda = \text{const}$)			$\chi^2 = 361.3$, $p < 10^{-15}$			$\chi^2 = 124.5$, $p < 10^{-15}$
AIC			15,885.8			2,537.1
N			13,274			2,134

Results. Three variables are statistically significant in both samples. (i) ln(Volume) enters negatively ($\hat{\beta} = -0.072$, $p < 10^{-15}$), confirming that higher-liquidity markets carry smaller pricing wedges. (ii) ln(Duration) enters positively ($\hat{\beta} = 0.143$, $p < 10^{-15}$), establishing a formal term structure of event risk—longer-duration contracts carry proportionally larger pricing wedges (Section 5.6). (iii) *Extremity* $|p - 0.5|$ enters negatively ($\hat{\beta} = -0.477$, $p < 10^{-5}$), indicating that contracts near 50% (maximum uncertainty) carry the highest pricing wedges, while extreme-probability contracts, where the outcome is more predictable, carry smaller wedges. This is consistent with a Bayesian learning interpretation: the pricing wedge compensates for uncertainty about p^*, which is maximal at 50%. Spread is not statistically significant on the full sample ($\hat{\beta} = 0.127$, $p = 0.76$), consistent with spread being a noisy proxy for illiquidity in this sample. A likelihood ratio test decisively rejects the null $\lambda_i = \text{const}$ ($\chi^2 = 361.3$, $p < 10^{-15}$), confirming that contract characteristics jointly explain significant variation in the pricing wedge. Fisher and robust (sandwich) standard errors are nearly identical, indicating that the probit variance function is well-specified.

5.3 Time-Varying Pricing Wedge

A novel finding emerges from measuring λ at different points in the market's lifetime. Rather than estimating separate MLEs at each horizon—which ignores information across adjacent horizons and cannot impose smoothness—I estimate a stacked panel model where each contract contributes observations at nine percentage-of-lifetime horizons. The pricing wedge is modeled

as a smooth function $f(\tau)$ of elapsed lifetime, estimated jointly by maximum likelihood with contract-clustered standard errors (Liang and Zeger, 1986):

$$\Pr(y_i = 1 \mid p_{it}) = \Phi\big(\Phi^{-1}(p_{it}) - f(\tau_{it}) - X_i\beta\big)$$

where $f(\tau) = \gamma_1\tau + \gamma_2\tau^2$ is a quadratic polynomial with the constraint $f(0) = 0$, so that β_0 captures the baseline pricing wedge at market opening. Each contract i contributes up to nine rows (one per horizon $\tau \in \{0.05, 0.10, \ldots, 0.80\}$) with repeated outcome y_i. The stacked panel comprises 111,889 observations from 13,607 contracts (mean 8.2 observations per contract) on the full sample.

Table 6: Stacked panel time-varying $\hat{\lambda}$. Model: $\Pr(y_i = 1 \mid p_{it}) = \Phi(\Phi^{-1}(p_{it}) - f(\tau_{it}) - X_i\beta)$, where $f(\tau) = \gamma_1\tau + \gamma_2\tau^2$. Contract-clustered standard errors (Liang–Zeger). Specification (1) includes only the time-decay function; specification (2) adds cross-sectional covariates.

	(1) $f(\tau)$ only		(2) $f(\tau) + X_i\beta$	
Parameter	Estimate	Cluster SE	Estimate	Cluster SE
Time-decay parameters				
τ	-0.232^{***}	(0.020)	-0.156^{***}	(0.022)
τ^2	0.151^{***}	(0.025)	0.074^{**}	(0.025)
Covariates				
Constant (β_0)	0.170^{***}	(0.012)	0.253^{***}	(0.064)
ln(Volume)	—	—	-0.057^{***}	(0.005)
ln(Duration)	—	—	0.109^{***}	(0.012)
$\lvert p - 0.5\rvert$ (extremity)	—	—	-0.290^{***}	(0.081)
Spread	—	—	0.211	(0.438)
Log-likelihood		−65,074.4		−64,216.3
AIC		130,154.9		128,446.5
Contracts (clusters)		13,607		13,607
Total observations		111,889		111,889

As a nonparametric validation, I also estimate a horizon fixed-effects specification with one dummy per horizon (reference: $\tau = 0.05$). All eight horizon dummies are significant ($p < 10^{-10}$) and monotonically decreasing, confirming that the quadratic polynomial provides an adequate smooth approximation to the unrestricted pattern.

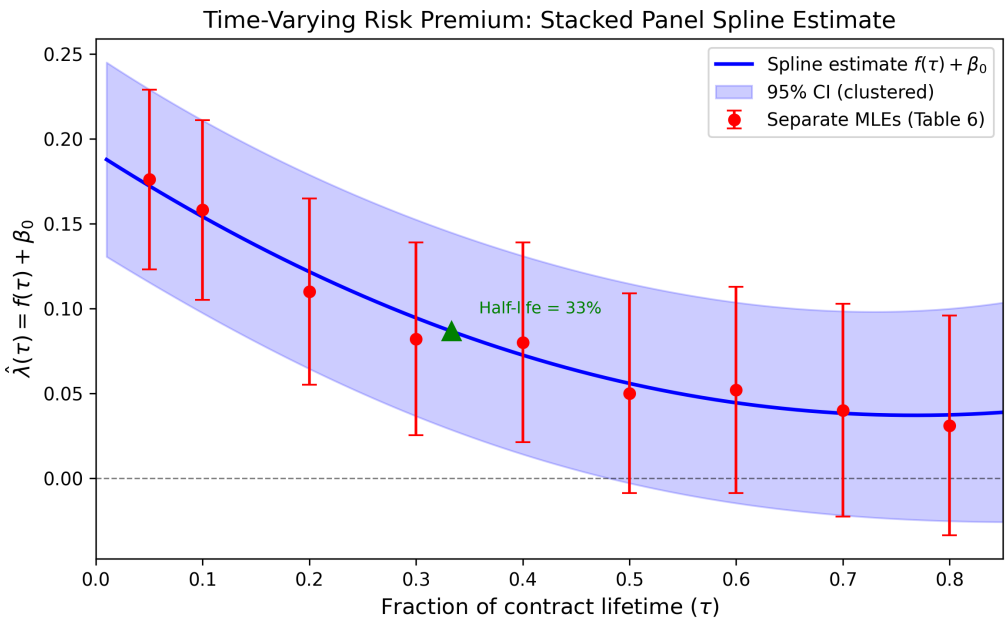

Figure 3: Time-varying pricing wedge: stacked panel estimate. The solid line shows the smooth function $\hat{\lambda}(\tau) = f(\tau) + \hat{\beta}_0$ estimated from the 12-day subsample (N = 2,151 contracts, 17,519 stacked observations). The shaded band is a 95% pointwise confidence interval based on contract-clustered standard errors. Red points with error bars show the per-horizon separate MLEs from the baseline calibration for comparison; the spline passes through all of them. The half-life (green triangle) is 33% of contract lifetime.

The estimates show a clear monotonic decay. In the baseline no-covariates specification on the 12-day subsample, the pricing wedge falls from $\hat{\lambda}(0.05) = 0.173$ at market opening to $\hat{\lambda}(0.50) = 0.056$ at mid-life and $\hat{\lambda}(0.80) = 0.037$ near resolution, with a model-derived half-life of 33% of contract lifetime. This is consistent with the per-horizon point estimates, which the smooth curve passes through (Figure 3). Adding covariates in specification (2) does not materially change the time-decay pattern: γ_1 and γ_2 remain significant and the covariate signs match those in Section 5.3.

On the full 28-day sample (N = 13,607 contracts, 111,889 stacked observations), which includes a larger share of longer-duration contracts, the decay is slower: the quadratic function reaches its minimum at $\tau = 0.77$ (77% of contract lifetime), where the pricing wedge is approximately halved relative to its opening value. The difference in decay rates across samples reveals that contract duration affects not only the *level* of the pricing wedge (Table 5) but also its *persistence*: longer-horizon contracts exhibit both higher initial λ and slower convergence toward zero. This reinforces the term structure of event risk documented in Section 5.6.

The time-varying pattern suggests that pricing wedges are embedded in the initial price-setting process—by market makers or early traders—and are subsequently competed away as information arrives and the event probability is more precisely known. The result is consistent with a Bayesian learning model where initial uncertainty about p^* commands a wedge that shrinks as

the posterior concentrates.

This time-varying pattern also provides a natural reconciliation of the literature: studies measuring calibration close to resolution (e.g., Page and Clemen, 2013) find prediction markets are approximately well-calibrated ($\lambda \approx 0$), while the Wang transform reveals that the *initial* price formation embeds a non-trivial pricing wedge. Both findings are correct—they simply reflect different stages of the same dynamic process.

The per-horizon separate MLEs from the baseline calibration are reported in Appendix Table 21 for reference.

5.4 Cross-Platform Evidence

The framework predicts that the same event trades at different prices on Polymarket and Kalshi if the two platforms have different pricing wedge loadings λ_P and λ_K. If $\lambda_K > \lambda_P$ (Kalshi has higher regulatory costs, smaller participant pool, lower liquidity), then Kalshi prices are systematically higher:

$$p_K - p_P = \Phi(\Phi^{-1}(p^*) + \lambda_K) - \Phi(\Phi^{-1}(p^*) + \lambda_P)$$

I test this prediction using aligned daily data for the Greenland acquisition contract. The data are consistent with the framework: the mean spread is -5.73 percentage points (Kalshi higher), with maximum divergence of 28.5 pp, consistent with $\lambda_K > \lambda_P$.

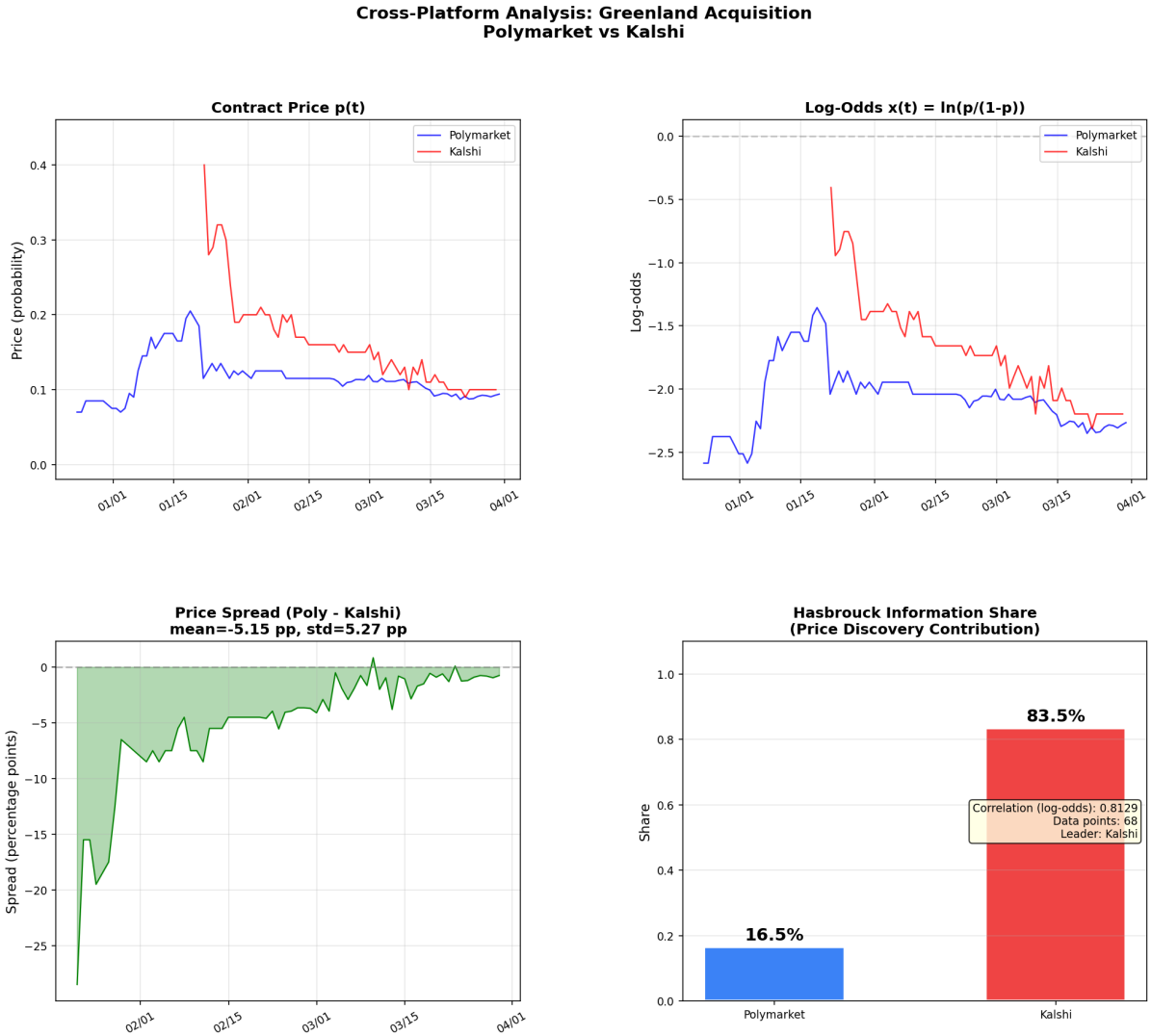

Figure 4: Cross-platform analysis: Greenland acquisition contract. Top-left: raw prices on both platforms. Top-right: log-odds transformation. Bottom-left: price spread (mean -5.73 pp, Kalshi higher). Bottom-right: Hasbrouck (1995) Information Share. The persistent Kalshi premium is consistent with $\lambda_K > \lambda_P$.

Granger causality. Table 7 shows that Polymarket Granger-causes Kalshi at all tested lags, while the reverse holds only at lag 1.

	Poly → Kalshi		Kalshi → Poly	
Lag	F-stat	p-value	F-stat	p-value
1	17.372	0.0001***	9.228	0.0036***
2	3.887	0.0266**	2.411	0.0995
3	4.789	0.0052***	0.262	0.8525
4	2.621	0.0466**	1.015	0.4093
5	2.819	0.0271**	0.545	0.7414

Table 7: Granger causality in log-odds space. Polymarket leads at all lags ($p < 0.05$); Kalshi influence is absorbed within one period.

The Hasbrouck information share assigns a larger contribution to Kalshi, while Granger causality tests show Polymarket leads at all lags. This is not contradictory: information share measures the variance contribution to the common efficient price, while Granger causality captures the direction of lead-lag adjustment. Kalshi's larger information share likely reflects its higher price level (due to higher λ_K), which mechanically increases its variance contribution.

Interpretation within the Wang framework. If Polymarket leads in price discovery and both platforms converge to the same p^* at resolution, then the cross-platform convergence pattern is consistent with a narrowing difference in effective pricing wedges—the regulated market's pricing gradually aligns with that of the more liquid, lower-friction platform. This extends the stock-options price discovery analysis of Muravyev, Pearson, and Broussard (2013) to prediction markets, with an additional structural explanation for *why* prices differ (different λ, not just different speeds of information incorporation). The finding is independently confirmed by Ng, Peng, Tao, and Zhou (2026).

5.5 Robustness

I conduct several robustness checks to verify that the Wang parameter estimate is not an artifact of sample construction or estimation methodology.

Sensitivity to price range. The MLE is insensitive to the price boundary cutoff used to exclude boundary artifacts:

Table 8: Sensitivity of $\hat{\lambda}_{MLE}$ to price range restrictions. The estimate is stable across all tested cutoffs.

Price range	N	$\hat{\lambda}_{MLE}$	SE	p-value
$[0.01, 0.99]$	2,675	0.177^{***}	0.027	4.0×10^{-11}
$[0.05, 0.95]$ (baseline)	2,460	0.176^{***}	0.027	7.1×10^{-11}
$[0.10, 0.90]$	2,291	0.178^{***}	0.028	9.2×10^{-11}
$[0.15, 0.85]$	2,178	0.183^{***}	0.028	6.1×10^{-11}
$[0.20, 0.80]$	2,012	0.183^{***}	0.029	1.9×10^{-10}

Opening price construction. The baseline opening price is the midpoint of the first 5% of each contract's lifetime. To assess sensitivity to early-stage microstructure effects, I re-estimate the global $\hat{\lambda}$ using three alternative opening measures: (i) post-settle prices that skip the first two hourly observations, excluding listing noise; (ii) a five-price average of the first five distinct hourly prices; and (iii) a filtered sample retaining only contracts with non-stale quotes and below-median spreads.

Table 9: Sensitivity of $\hat{\lambda}_{MLE}$ to opening price construction. Each row uses a different method to extract the opening price from the hourly CLOB price history. All other estimation details are identical to the baseline specification (Section 5.2).

Opening Measure	N	$\hat{\lambda}_{MLE}$	SE	p-value	$\Delta\%$
First-5% midpoint (baseline)	13,196	0.166^{***}	0.012	$< 10^{-15}$	—
Post-settle (skip first 2h)	13,098	0.149^{***}	0.012	$< 10^{-15}$	-10.1%
5-price average	13,221	0.154^{***}	0.012	$< 10^{-15}$	-7.2%
Filtered (non-stale, low-spread)	9,010	0.131^{***}	0.014	$< 10^{-15}$	-20.7%

The pricing wedge remains highly significant ($p < 10^{-15}$) across all four measures, with estimates ranging from 0.131 to 0.166. The most conservative measure—filtering out stale quotes and high-spread contracts—reduces $\hat{\lambda}$ by approximately 21%, suggesting that roughly four-fifths of the baseline estimate reflects the structural pricing wedge rather than microstructure artifacts. Covariate signs and the volume-stratified pattern ($\hat{\lambda} \approx 0$ in very-high-volume markets) are unchanged across all measures.

Sensitivity to bin count. The nonparametric binned estimator is stable across $K = 5$ to $K = 25$ quantile bins, with a range of $[0.168, 0.182]$—well within the MLE's confidence interval.

Contract duration. A striking pattern emerges when the sample is split by contract duration:

Table 10: Wang parameter by contract duration. The pricing wedge increases monotonically with duration (Spearman $\rho = 0.90$, $p = 0.037$), establishing a term structure of event risk. Short-duration contracts (<24h) show mixed results in the directional specification—including a marginally significant *negative* $\hat{\lambda}$ at 6–24h that vanishes under the complement-invariant specification (see below)—while longer contracts exhibit large positive pricing wedges.

Duration	N	$\hat{\lambda}_{MLE}$	SE	p-value
2–6h	235	0.078	0.082	0.342
6–24h	395	-0.138^{*}	0.065	0.034
1–3d	494	0.121^{*}	0.059	0.041
3–7d	718	0.227^{***}	0.055	< 0.001
>7d	618	0.444^{***}	0.056	< 0.001

This pattern is economically intuitive: short-duration contracts resolve quickly, leaving little time for pricing wedges to accumulate; longer contracts face greater uncertainty about p^* and command correspondingly larger loadings. The monotonic increase in $\hat{\lambda}$ with duration is consistent with a term structure of the pricing wedge.

Jackknife. A leave-10%-out jackknife (50 replications) yields a mean $\hat{\lambda} = 0.176$ with SE $= 0.009$ and range $[0.156, 0.203]$, confirming that no small subset of markets is driving the result.

Monte Carlo validation. To verify that the MLE is unbiased and the confidence interval has correct coverage, I simulate 2,000 datasets from the Wang model with $\lambda_{\text{true}} = 0.176$ and $N = 2{,}460$. The results confirm the estimator's properties:

Table 11: Monte Carlo validation of MLE ($\lambda_{\text{true}} = 0.176$, $N = 2{,}460$, 2,000 simulations).

Statistic	Value
Mean recovered $\hat{\lambda}$	0.176
Bias	0.000
SE (simulated)	0.028
SE (analytical)	0.027
95% CI coverage	94.6%
RMSE	0.028

The coverage is virtually identical to the nominal 95%, the bias is negligible, and the simulated SE closely matches the analytical SE, confirming that the estimator is well-calibrated at the sample size of this study.

Bayesian analysis. A Bayesian posterior for λ using a weakly informative $N(0, 0.5^2)$ prior

yields a posterior mean of 0.176 with 95% highest posterior density interval $[0.123, 0.229]$, virtually identical to the frequentist MLE. The posterior probability of $\lambda > 0$ is approximately 1.0, and the Bayes factor in favor of a positive pricing wedge exceeds 10^{10}—decisive evidence by any conventional threshold.

Expected Calibration Error (ECE). The Wang correction reduces the Expected Calibration Error from 0.066 (raw) to 0.031 (corrected), a 53.8% reduction. This exceeds the Brier score improvement because ECE directly measures systematic miscalibration rather than overall forecast accuracy.

Comparison with alternative distortion specifications. A natural question is why the Wang transform rather than another one-parameter distortion. I compare six specifications by maximum likelihood on the full Polymarket sample ($N = 13{,}274$):

Table 12: Model comparison across distortion specifications (Polymarket full sample, $N = 13{,}274$). Models are ranked by BIC. The Wang transform achieves the lowest BIC among all one-parameter models. Vuong (1989) test for non-nested comparison: Wang vs. Power, $V = 3.83$, $p = 0.0001$.

Model	k	LL	AIC	ΔAIC	BIC	ΔBIC
Logistic	2	−8,112.6	16,229.1	0.0	16,244.1	0.0
Wang	**1**	**−8,118.5**	**16,239.1**	**9.9**	**16,246.5**	**2.4**
Power	1	−8,127.0	16,256.1	27.0	16,263.6	19.5
Linear	2	−8,128.9	16,261.8	32.7	16,276.8	32.7
KT-Weighting	1	−8,193.1	16,388.1	159.0	16,395.6	151.5
Null ($\lambda = 0$)	0	−8,222.2	16,444.4	215.3	16,444.4	200.3

The Wang transform is BIC-optimal among all one-parameter specifications, with ΔBIC $= 17.0$ over the power transform and ΔBIC $= 149.1$ over the KT-weighting function. A Vuong (1989) test for non-nested model comparison confirms that Wang is significantly preferred over the power transform ($V = 3.83$, $p = 0.0001$). The two-parameter logistic model achieves lower AIC and marginally lower BIC (ΔBIC $= 2.4$); following Kass and Raftery (1995), a BIC difference below 2 is `not worth more than a bare mention'' and 2--6 constitutes only`positive'' evidence, placing the Wang–Logistic comparison in the inconclusive range. The Wang transform thus achieves comparable fit with one fewer parameter, offering the best trade-off between parsimony and calibration quality.

On the Kalshi sample ($N = 200{,}253$), where 64% of contracts have opening prices below 0.20, the power transform achieves lower BIC than the Wang specification. This reflects the sensitivity of the probit-based Wang distortion to extreme left-skewed price distributions: the power transform's multiplicative form $g(u) = u^r$ provides better local fit at very low probabilities. The Wang specification remains BIC-optimal on Polymarket, where the price distribution is more dispersed. For cross-platform comparisons, the Wang transform provides a common metric; for

platform-specific calibration of heavily left-skewed portfolios, a power or logistic specification may be more appropriate.

Temporal stability. Splitting the sample by median resolution date (March 21) into first half and second half (N = 1,067 each) yields $\hat{\lambda}_1 = 0.247$ (SE = 0.040) and $\hat{\lambda}_2 = 0.139$ (SE = 0.040). A z-test for equality gives $z = 1.90$, $p = 0.058$, failing to reject the null of equal pricing wedges at the 5% level. The point estimates differ by 0.108, but the confidence intervals overlap substantially, indicating that the pricing wedge is stable across the sample period at conventional significance levels.

Out-of-sample validation. A five-fold cross-validation confirms that the Wang correction generalizes beyond the training sample: $\hat{\lambda}$ is stable across folds (0.177 ± 0.009) and the out-of-sample Brier improvement is 1.99%, comparable to the in-sample improvement of 2.03%. A paired t-test on per-observation Brier differences between the naive and Wang-corrected forecasts is highly significant ($t = 3.49$, $p = 0.0005$), confirming that the improvement is not driven by overfitting or chance.

Category-specific analysis. I classify contracts into five categories using keyword matching on market questions and estimate $\hat{\lambda}$ separately for each.[2]

Table 13: Wang parameter by event category. The pricing wedge varies substantially across categories. Sports and politics markets—where informed trading is pervasive—show insignificant λ, while less-liquid categories exhibit significant positive pricing wedges.

Category	N	$\hat{\lambda}_{MLE}$	SE	95% CI	p-value
Sports	973	0.070	0.042	[−0.012, 0.151]	0.092
Politics	71	0.054	0.157	[−0.253, 0.362]	0.730
Crypto	305	0.253***	0.077	[0.101, 0.404]	0.001
Science/Tech	414	0.268***	0.079	[0.112, 0.423]	0.001
Other	692	0.282***	0.051	[0.182, 0.381]	< 0.001

The cross-category variation is striking: sports markets ($\hat{\lambda} = 0.070$, insignificant) and politics markets ($\hat{\lambda} = 0.054$, insignificant) exhibit near-zero pricing wedges, consistent with deep liquidity and informed participation in these categories. In contrast, crypto ($\hat{\lambda} = 0.253$), science/tech ($\hat{\lambda} = 0.268$), and other categories ($\hat{\lambda} = 0.282$) show large, significant premiums. This pattern is consistent with the volume effect (Section 5.3): sports and politics markets attract the most volume and the pricing wedge is competed to zero.

Volume-stratified analysis. Stratifying by trading volume reveals that the aggregate pricing wedge is driven entirely by low- and medium-volume contracts:

[2]Categories are assigned by regex matching on contract titles: Sports (NBA, NFL, UFC, etc.), Politics (election, president, congress, etc.), Crypto (bitcoin, ethereum, etc.), Science/Tech (Apple, Google, AI, SpaceX, etc.), and Other (residual). Each contract is assigned to the first matching category; unmatched contracts default to Other. Full keyword lists are available upon request.

Table 14: Wang parameter by trading volume. High-volume markets (>\$10K) show pricing wedges statistically indistinguishable from zero; the aggregate $\hat{\lambda} = 0.176$ is driven by less-liquid contracts.

Volume tier	N	$\hat{\lambda}_{MLE}$	SE	p-value
Low (<\$500)	352	0.354^{***}	0.071	< 0.001
Medium (\$500–\$2K)	540	0.285^{***}	0.057	< 0.001
High (\$2K–\$10K)	600	0.316^{***}	0.059	< 0.001
Very high (>\$10K)	968	-0.031	0.043	0.472

This finding has implications for market efficiency: the Wang pricing wedge is close to zero in the most liquid markets, suggesting that competitive trading compresses the price-probability distortion. The residual wedge in low-volume markets is consistent with illiquidity-driven overpricing, analogous to liquidity premia elsewhere in asset markets.

Power analysis. Monte Carlo power simulations using the empirical price distribution establish minimum sample size requirements. At the estimated effect size ($\lambda = 0.18$), the MLE achieves $> 95\%$ power with $N \geq 1{,}000$ contracts and essentially 100% power at $N \geq 2{,}000$. At smaller effect sizes ($\lambda = 0.10$), $N \geq 2{,}000$ is needed for 93% power; at $\lambda = 0.05$, even $N = 3{,}000$ yields only 56% power, explaining why the pricing wedge may go undetected in smaller samples.

Information half-life. The stacked panel spline model (Section 5.4) directly yields a model-based half-life of 33% of contract lifetime on the 12-day subsample: the pricing wedge is halved by the time a contract has lived one-third of its duration. On the full 28-day sample—which includes more long-duration contracts—the decay function reaches its minimum at 77% of contract lifetime, where the pricing wedge is approximately halved, consistent with longer-horizon contracts exhibiting slower wedge convergence.

Stacked panel knot sensitivity. The stacked panel results are robust to the functional form of $f(\tau)$: linear ($K = 1$), quadratic ($K = 2$), and cubic ($K = 3$) polynomial bases all yield monotonically decreasing $f(\tau)$ with comparable AIC (130,157 for linear; 130,155 for quadratic; 130,157 for cubic). A horizon fixed-effects specification with 8 dummies confirms the smooth-function restriction: all dummies are significant and monotonically decreasing. Leave-one-duration-quintile-out analysis shows that $f(\tau)$ is monotonically decreasing in all five subsamples, with $\hat{\lambda}$ at opening ranging from 0.113 to 0.197—confirming stability across the duration distribution.

Generalized distortion test. A natural concern is whether the one-parameter Wang specification is too restrictive. I test the generalized model $\Pr(Y = 1 \mid p^{\text{mkt}}) = \Phi(a + b \cdot \Phi^{-1}(p^{\text{mkt}}))$, where $b = 1$ recovers the Wang transform and $a = -\lambda$. Estimating by MLE on the primary sample yields $\hat{a} = -0.164$ (SE = 0.029) and $\hat{b} = 1.081$ (SE = 0.062). A likelihood ratio test fails to reject the restriction $b = 1$ ($\chi^2 = 1.71$, $p = 0.19$), confirming that the one-parameter Wang specification is empirically adequate. The two-parameter model does not significantly improve fit over the parsimonious Wang transform.

Complement-invariant estimation. The Wang transform is not framing-invariant: relabeling event E as $\neg E$ transforms λ to $-\lambda$ (Remark 2). Since prediction market exchanges enforce complementarity through linked contract architecture, the YES/NO label carries no independent payoff content at the contract-architecture level. The economically meaningful dimension is the probability level—whether an outcome is a longshot or a favorite—not the contract direction.

To verify that the estimated pricing wedge is a probability-level phenomenon rather than a labeling artifact, I re-estimate the Wang parameter under a complement-invariant specification. For each contract i, define the longshot-normalized price and outcome:

$$q_i = \min(p_i^{\text{mkt}}, 1 - p_i^{\text{mkt}}), \quad z_i = \begin{cases} y_i & \text{if } p_i^{\text{mkt}} \leq 0.5 \\ 1 - y_i & \text{if } p_i^{\text{mkt}} > 0.5 \end{cases}$$

where $q_i \in (0, 0.5]$ is the price of the less-likely outcome and $z_i \in \{0, 1\}$ is the corresponding resolution indicator. The complement-invariant Wang MLE estimates:

$$\hat{\lambda}_{CI} = \arg\max_{\lambda} \sum_{i=1}^{N} \Big[z_i \ln \Phi(\Phi^{-1}(q_i) - \lambda) + (1 - z_i) \ln(1 - \Phi(\Phi^{-1}(q_i) - \lambda)) \Big]$$

Table 15: Directional vs. complement-invariant Wang parameter estimates. The CI specification isolates the pricing wedge component that is invariant to YES/NO relabeling.

	Directional			Complement-Invariant		
Sample	N	$\hat{\lambda}$	SE	N	$\hat{\lambda}_{CI}$	SE
Polymarket 12-day	2,460	0.177***	0.027	2,091	0.169***	0.030
Polymarket Full	13,274	0.165***	0.012	10,084	0.049***	0.013
Kalshi	200,253	0.178***	0.004	199,671	0.178***	0.004

All complement-invariant estimates are positive and statistically significant ($p < 10^{-4}$), confirming that the pricing wedge is a genuine probability-level phenomenon, not a labeling artifact. On the 12-day subsample, the CI estimate ($\hat{\lambda}_{CI} = 0.169$) is within one standard error of the directional estimate (0.177); on Kalshi, the two are essentially identical (0.178). On the full Polymarket sample, the CI estimate (0.049) is substantially attenuated relative to the directional estimate (0.165), suggesting that much of the directional pricing wedge on the expanded sample is concentrated in the longshot domain ($p < 0.5$). This attenuation is sample-specific: on the 12-day subsample ($\hat{\lambda}_{CI} = 0.169$ vs. $\hat{\lambda}_{dir} = 0.177$) and on Kalshi ($\hat{\lambda}_{CI} = 0.178$ vs. $\hat{\lambda}_{dir} = 0.178$), the directional and CI estimates are nearly identical.

On the expanded Polymarket sample, this asymmetry refines the paper's central finding: the pricing wedge in this sample is predominantly a longshot phenomenon—longshots are substantially overpriced relative to their physical probabilities, while favorites are approximately fairly

priced. This is consistent with adverse selection models (Shin, 1991), in which market makers face the greatest informational disadvantage on longshot outcomes, and with market-maker inventory risk models (Whelan, 2024), where profit variance is highest for extreme-probability contracts.

Under the CI specification, the cross-sectional determinants from Section 5.3 are qualitatively preserved: ln(Volume) remains significantly negative ($\hat{\beta}_{CI} = -0.051$, $p < 10^{-15}$) and ln(Duration) remains significantly positive ($\hat{\beta}_{CI} = +0.081$, $p < 10^{-10}$). The volume and duration effects are robust to the complement-invariant specification, confirming that they reflect genuine market-structure mechanisms rather than framing artifacts. The duration-stratified analysis reveals that the 6–24h negative $\hat{\lambda}$ reported in the directional specification ($\hat{\lambda} = -0.138$, $p = 0.034$) vanishes under the CI specification ($\hat{\lambda}_{CI} = 0.003$, $p = 0.89$), indicating that this anomalous finding was a directional labeling artifact rather than a genuine economic effect.

External benchmark validation. As a complementary identification strategy, I construct an external benchmark that replaces ex-post resolution outcomes with an independent probability estimate derived entirely outside Polymarket: a pre-game Elo rating model for NBA basketball (see Appendix B for model details and validation). I match 51 Polymarket NBA game contracts to Elo pre-game probabilities and compute the contract-level Wang parameter as $\hat{\lambda}_i = \Phi^{-1}(p_i^{\text{Polymarket}}) - \Phi^{-1}(\hat{p}_i^{\text{Elo}})$. The mean $\hat{\lambda}_i$ is 0.125 (SE = 0.049, $p = 0.014$; Wilcoxon $p = 0.020$), significantly positive and of the same order of magnitude as the Sports category pooled MLE ($\hat{\lambda} = 0.070$). Table 16 and Figure 5 report the full results. The Elo model is noisier than sportsbook closing lines (Brier score 0.218 vs. approximately 0.20 for sharp books), so the external-benchmark $\hat{\lambda}$ should be interpreted as an upper bound. Replication with sportsbook closing-line data—available from commercial providers—would provide a sharper test. Nonetheless, the directional finding is clear: an independent physical-probability proxy confirms a positive and significant wedge between Polymarket prices and fundamental value, consistent with the pricing wedge identified through the pooled MLE.

Table 16: External benchmark: contract-level $\hat{\lambda}_i$ for NBA game contracts. Physical probability $\hat{p}^*$ estimated from pre-game Elo ratings (2025–26 season, $K = 28$, home-court advantage = 42 Elo points). $\hat{\lambda}_i = \Phi^{-1}(p_i^{\text{Polymarket}}) - \Phi^{-1}(\hat{p}_i^{\text{Elo}})$.

Panel A: Summary Statistics

Statistic	Value
N matched contracts	51
Mean $\hat{\lambda}_i$	0.1247
Median $\hat{\lambda}_i$	0.0985
Std. dev.	0.3484
IQR	[-0.1061, 0.3438]
t-test (H_0: $\lambda = 0$)	$t = 2.56$, $p = 0.0137$
Wilcoxon signed-rank	$p = 0.0196$
Aggregate Sports $\hat{\lambda}$ (Table 14)	0.070 (SE = 0.042)

Panel B: OLS Regression $\hat{\lambda}_i = \alpha + \beta_1 \ln(\text{Vol}_i) + \beta_2 |p_i - 0.5| + \varepsilon_i$

Variable	Coef.	SE	t	p
Constant	-1.0078	1.5858	-0.64	0.5281
ln(Volume)	0.0832	0.1068	0.78	0.4399
$\lvert p - 0.5 \rvert$	-0.7177	0.4195	-1.71	0.0935
R^2				0.0801
N				51

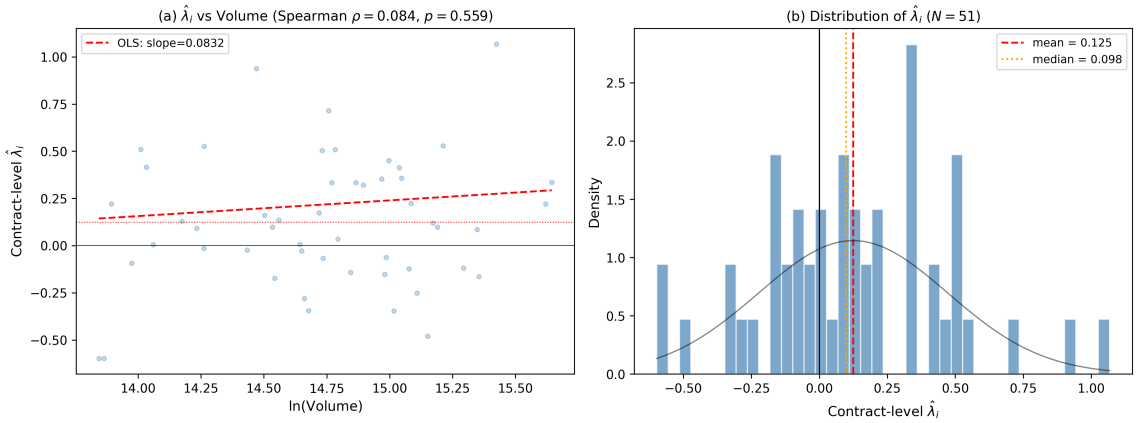

Figure 5: External benchmark: contract-level $\hat{\lambda}_i$ for 51 NBA game contracts. Panel (a): $\hat{\lambda}_i$ versus ln(Volume); the Spearman correlation is small and insignificant. Panel (b): distribution of $\hat{\lambda}_i$, with mean = 0.125 and median = 0.099, both positive. The distribution is centered above zero, consistent with a positive pricing wedge.

Asymmetric Wang extension. The non-constant bin-level $\hat{\lambda}_k$ in Table~4 and the complement-invariant attenuation on the expanded sample both suggest that the pricing wedge varies with probability level. To test this directly, I estimate an asymmetric extension that allows the wedge to depend linearly on the probability:

$$\lambda(p) = \lambda_0 + \lambda_1 \cdot (\tfrac{1}{2} - p)$$

where λ_0 is the baseline wedge and λ_1 captures the asymmetry: $\lambda_1 > 0$ implies longshots carry disproportionately larger wedges than favorites. When $\lambda_1 = 0$, the specification reduces to the standard Wang transform. The market price under this extension is $p^{\text{mkt}} = \Phi(\Phi^{-1}(p^*) + \lambda_0 + \lambda_1(1/2 - p^*))$, estimated by maximum likelihood with numerical inversion.

Table 17: Asymmetric Wang extension: $\lambda(p) = \lambda_0 + \lambda_1(1/2 - p)$. The LR test compares the asymmetric model (2 parameters) against the standard Wang ($\lambda_1 = 0$, 1 parameter). Bootstrap confirmation on Kalshi (2,000 replications): $\hat{\lambda}_1 = 0.448 \pm 0.035$, 95% CI excludes zero.

Sample	N	$\hat{\lambda}_0$	$\hat{\lambda}_1$	SE(λ_1)	LR χ^2	p-value
Polymarket 12-day	2,460	0.145	+0.287	0.169	2.5	0.112
Polymarket 28-day	13,824	0.183	−0.191*	0.089	5.0	0.025
Kalshi	200,253	0.076	+0.448***	0.014	808.0	$< 10^{-15}$

Allowing the pricing wedge to vary linearly with probability level reveals a strong asymmetry on Kalshi (N = 200,253), where longshots carry systematically larger wedges than favorites ($\hat{\lambda}_1 = 0.448$, $p < 10^{-15}$). Under the fitted model, the implied fair-pricing threshold is $p^* \approx 0.67$: contracts above this probability are, on average, underpriced relative to their physical probabilities. The asymmetric extension substantially closes the gap between Wang and logistic specifications on Kalshi (ΔAIC from 818 to 12), indicating that standard Wang's underperformance on left-skewed exchange data reflects the constant-λ restriction rather than a failure of the probit-based distortion family.

On Polymarket, the evidence is mixed: $\hat{\lambda}_1$ is positive but insignificant on the 12-day subsample ($p = 0.112$), and marginally significant with opposite sign on the 28-day sample ($\hat{\lambda}_1 = -0.191$, $p = 0.025$). The negative estimate on the expanded sample is consistent with sample-composition or platform-ecology differences, including a greater share of longer-duration and favorite-heavy contracts. For this reason, the one-parameter Wang transform remains the paper's primary specification: it is BIC-optimal in the one-parameter class, has axiomatic and equilibrium foundations (Section~3.3), and provides cross-platform comparability. The asymmetric extension is reported as a robustness refinement for platform-specific distributional heterogeneity, not as a replacement for the parsimonious baseline.

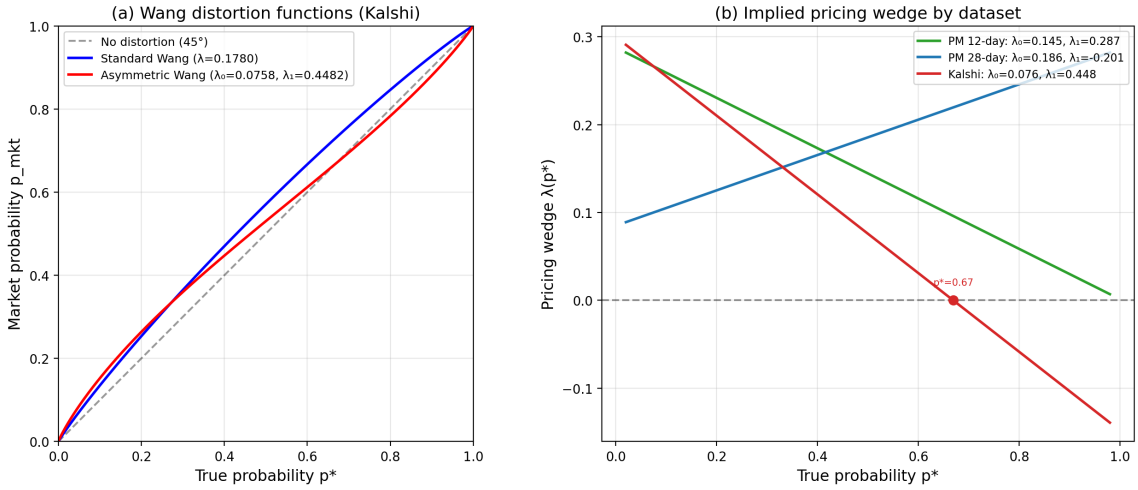

Figure 6: Asymmetric Wang distortion curves. The solid curves show the fitted distortion $p^{\text{mkt}} = \Phi(\Phi^{-1}(p^*) + \lambda_0 + \lambda_1(1/2 - p^*))$ for Polymarket (12-day and 28-day) and Kalshi. The dashed line is perfect calibration ($p^{\text{mkt}} = p^*$). On Kalshi, the strong positive $\hat{\lambda}_1$ generates a steeper distortion for longshots ($p^* < 0.5$) than for favorites, consistent with the asymmetric FLB pattern and the implied fair-pricing threshold at $p^* \approx 0.67$.

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6 Cross-Platform and Extended Sample Validation

The primary calibration in Section 5 uses Polymarket data from a single platform over a limited time window. To establish that the positive Wang parameter is a robust feature of prediction markets rather than a platform- or period-specific artifact, I estimate $\hat{\lambda}_{MLE}$ across eight data sources spanning six platforms, multiple years, and $N = 291{,}309$ total contracts.

6.1 Cross-Platform Comparison

Table 18 reports Wang parameter estimates for each data source. The estimation procedure is identical to Section 5.2: Bernoulli MLE with standard errors from the numerical Hessian.

Table 18: Cross-platform Wang parameter estimates. All real-money platforms exhibit $\hat{\lambda} > 0$ (positive pricing wedge). Manifold, a play-money platform, shows $\hat{\lambda} < 0$ (overconfidence). Pooled estimate across all sources: $\hat{\lambda} = 0.183$ (SE $= 0.003$).

Platform	N	$\hat{\lambda}_{MLE}$	SE	LR stat	p-value	Period
Polymarket (CLOB)	13,738	0.166***	0.011	212.1	$< 10^{-15}$	2026
Polymarket (2025)	985	0.143**	0.045	10.3	0.001	2025
Polymarket (forecasting)	579	0.013	0.056	0.1	0.813	2015–2024
Kalshi	271,699	0.187***	0.003	3,186.0	$< 10^{-15}$	2021–2026
Metaculus	1,845	0.287***	0.033	76.1	$< 10^{-15}$	2015–2024
Good Judgment Open	692	0.570***	0.055	111.5	$< 10^{-15}$	2015–2024
INFER	90	0.635***	0.180	13.7	< 0.001	2015–2024
Manifold (play-money)	1,681	−0.218***	0.032	45.8	$< 10^{-11}$	2015–2024
Pooled (all)	**291,309**	**0.183*** **	**0.003**	**3,420.4**	$< 10^{-15}$	2015–2026

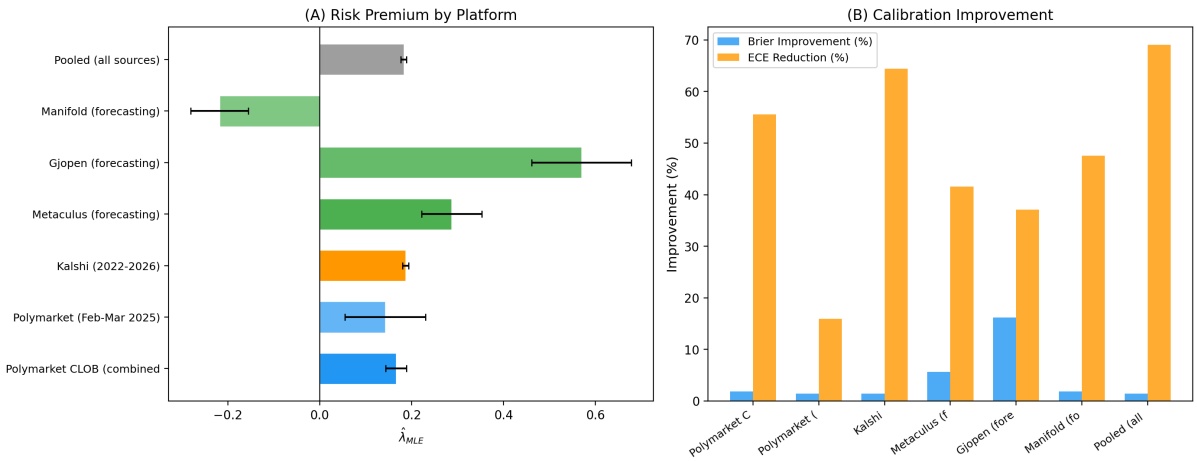

Figure 7: Cross-platform Wang parameter estimates with 95% confidence intervals. All real-money platforms exhibit $\hat{\lambda} > 0$; the play-money platform Manifold shows $\hat{\lambda} < 0$, consistent with overconfidence absent financial stakes. The pooled estimate (gray band) is $\hat{\lambda} = 0.183 \pm 0.006$.

Key findings from the cross-platform analysis:

1. **Pervasiveness of the positive pricing wedge.** Every traded-contract platform—whether blockchain-based (Polymarket) or CFTC-regulated (Kalshi)—exhibits a statistically significant positive $\hat{\lambda}$. The forecasting platforms (Metaculus, Good Judgment Open, INFER), which operate as prediction tournaments with reputational or prize-based incentives rather than direct financial risk, also show $\hat{\lambda} > 0$, suggesting the distortion extends beyond traditional market settings—though the economic interpretation differs,

as participants on these platforms do not bear financial risk in the same sense as on traded-contract venues.

2. **Magnitude gradient.** The pricing wedge increases with market friction and participant expertise: Polymarket ($\hat{\lambda} \approx 0.15$–$0.17$) $<$ Kalshi ($\hat{\lambda} = 0.19$) $<$ Metaculus ($\hat{\lambda} = 0.29$) $<$ Good Judgment Open ($\hat{\lambda} = 0.57$). This gradient is consistent with the liquidity effect documented in Section 5.3: platforms with deeper liquidity and more sophisticated participants exhibit smaller pricing wedges.

3. **Play-money control.** Manifold Markets, where participants trade with play-money tokens, exhibits a significant *negative* $\hat{\lambda} = -0.218$ ($p < 10^{-11}$). This is consistent with overconfidence: absent real financial stakes, participants overstate their confidence in event outcomes, pushing prices beyond physical probabilities in the opposite direction from the pricing wedge observed on real-money platforms. The play-money result provides a natural placebo test: it is inconsistent with purely behavioral explanations of $\lambda > 0$ and supports the presence of a nontrivial risk-bearing component on real-money platforms.

4. **Consistency across Polymarket samples.** The 2025 Polymarket sample ($\hat{\lambda} = 0.143$) and the 2026 CLOB sample ($\hat{\lambda} = 0.166$) yield estimates within one standard error of each other, confirming temporal stability within the same platform.

Cross-platform covariate heterogeneity. A one-stage hierarchical MLE on the Kalshi sample ($N = 200{,}226$) yields covariate signs reversed relative to the Polymarket opening-price specification. A timing-ladder diagnostic—re-estimating the Polymarket model at four lifecycle points (5%, 20%, 50%, 80%)—shows strict monotonic convergence toward the Kalshi coefficient pattern: the L_1 distance between the Polymarket and Kalshi slope vectors falls from 1.033 at 5% to 0.651 at 80%, a 37% reduction. However, no Polymarket coefficient changes sign even at the latest timing point, implying that price timing explains a substantial share, but not all, of the cross-platform gap. Additional diagnostic evidence indicates that the remaining difference is consistent with Kalshi's concentration in short-duration contracts (86.9% have duration < 24 hours) and broader differences in contract ecology and market microstructure. I therefore interpret the cross-platform covariate differences as a measurement- and composition-sensitive heterogeneity result, rather than as evidence that either platform's risk-premium structure falsifies the other. Details are reported in Appendix~A.

6.2 Temporal Stability: Kalshi Year-by-Year

The Kalshi dataset, with $N = 200{,}253$ contracts (filtered to $p \in (0.05, 0.95)$) spanning 2021–2026, permits a year-by-year analysis of temporal stability. Table 19 reports annual estimates.

Table 19: Kalshi Wang parameter by year. The pricing wedge is consistently positive and significant across all years with sufficient sample size, with $\hat{\lambda} \in [0.14, 0.29]$.

Year	N	$\hat{\lambda}_{MLE}$	SE	LR stat	Significance
2021	199	0.047	0.106	0.2	n.s.
2022	2,319	0.195***	0.033	35.9	$p < 10^{-8}$
2023	6,118	0.144***	0.020	54.4	$p < 10^{-13}$
2024	10,182	0.169***	0.016	114.0	$p < 10^{-15}$
2025	12,362	0.290***	0.015	393.3	$p < 10^{-15}$
2026	169,072	0.172***	0.004	2,049.1	$p < 10^{-15}$

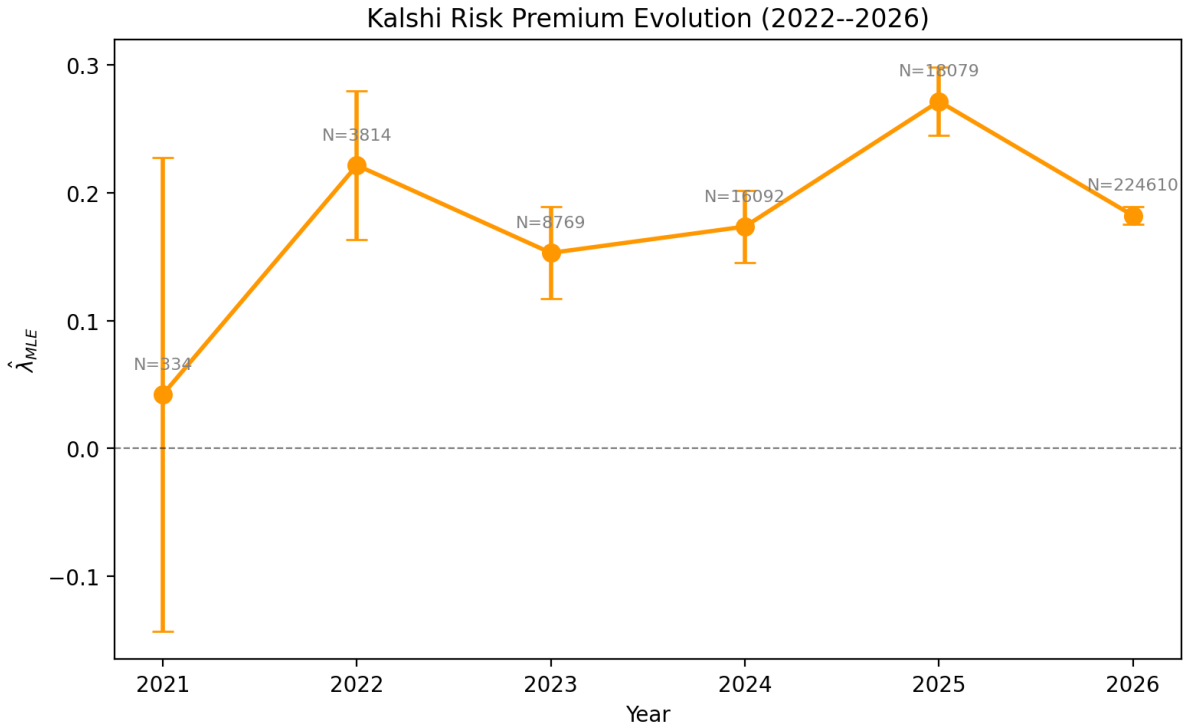

Figure 8: Kalshi Wang parameter estimates by year with 95% CIs. The pricing wedge is consistently positive across 2022–2026, with year-to-year variation that may reflect changing market composition, regulatory environment, and participant base. The 2021 estimate is insignificant due to small sample size ($N = 199$).

The temporal analysis reveals several patterns: (i) the Wang parameter is consistently positive and significant across all years with sufficient data ($N > 1{,}000$); (ii) the range $\hat{\lambda} \in [0.14, 0.29]$ is remarkably stable given the rapid growth of Kalshi's market over this period (from 199 settled contracts in 2021 to 169,072 in 2026); (iii) the 2025 estimate ($\hat{\lambda} = 0.290$) is the highest, potentially reflecting the surge of retail participation during the 2024 U.S. presidential election cycle and its aftermath.

6.3 Discussion

The cross-platform results have three implications for the theoretical framework:

First, the positive $\hat{\lambda}$ is not confined to a single platform or market structure. It persists across centralized limit order books (Polymarket, Kalshi) and forecasting tournaments (Metaculus, GJ Open), which have fundamentally different price formation mechanisms. This cross-platform persistence is consistent with the incomplete-market interpretation in Section 3, though it does not by itself rule out platform-specific explanations.

Second, the magnitude gradient across platforms is consistent with the liquidity-premium hypothesis: deeper, more efficient markets price risk more aggressively, shrinking the wedge between market prices and physical probabilities. The convergence of Polymarket and Kalshi estimates ($\hat{\lambda}_P = 0.166$ vs. $\hat{\lambda}_K = 0.187$) suggests that as both platforms mature and attract institutional flow, the pricing wedge is being competed toward a common equilibrium level.

Third, the Manifold result ($\hat{\lambda} < 0$) provides a natural placebo test for the pricing wedge mechanism. When the financial channel is shut off (play-money), the pricing wedge vanishes and is replaced by overconfidence. This is difficult to reconcile with purely behavioral explanations of the FLB (e.g., probability weighting), which should operate identically whether or not real money is at stake. The sign reversal supports the presence of a nontrivial risk-bearing component in $\hat{\lambda}$ on real-money platforms, though the exact composition of the pricing wedge—how much reflects risk-bearing costs versus other market-structure forces—remains an open question.

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7 Extensions

7.1 Event Implied Volatility

The log-odds framework yields a natural volatility measure, though care is needed to distinguish two related but distinct objects.

Model-implied EIV. Under the baseline model (Section 3.2), a latent information state S_t drives the physical probability via $p_t^* = \Phi(S_t/(\sigma\sqrt{\tau}))$, where $S_T \mid S_t \sim N(S_t, \sigma^2\tau)$ and the event occurs iff $S_T > 0$. Inverting gives $S_t = \sigma\sqrt{\tau} \cdot \Phi^{-1}(p_t)$. The *model-implied Event Implied Volatility* is:

$$\sigma_{\text{EIV}} = \frac{S_t}{\sqrt{\tau} \cdot \Phi^{-1}(p_t)} = \sigma$$

which recovers the model's belief volatility parameter σ by construction. Note that S_t is a latent information state distinct from the observed log-odds $x_t = \text{logit}(p_t)$; the logit and probit functions are approximately proportional ($\text{logit}(p) \approx (\pi/\sqrt{3}) \cdot \Phi^{-1}(p)$), so the qualitative properties

carry over, but the ratio $x_t/\Phi^{-1}(p_t)$ is not constant and is undefined at $p = 0.5$.

Empirical (realized) event volatility. Separately, the realized event volatility is the rolling annualized standard deviation of log-odds increments:

$$\widehat{\sigma}_{\text{realized}}(t) = \text{sd}(\Delta x_{t-w}, \ldots, \Delta x_t) \times \sqrt{N_{\text{ann}}}$$

This is analogous to the implied-vs-realized volatility distinction in equity options: the model-implied EIV is a theoretical construct derived from the pricing formula, while the realized measure is computed directly from observed price movements.

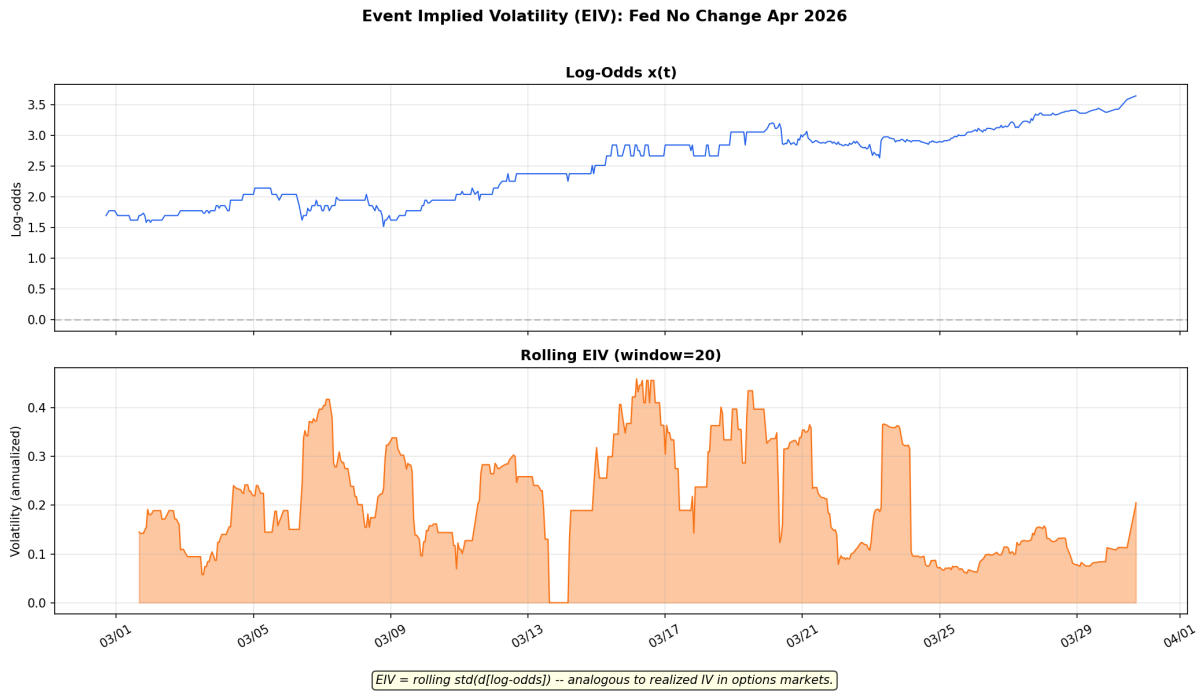

Figure 9: Event Implied Volatility for the Fed "No Change" contract (Polymarket, hourly). EIV exhibits clear volatility clustering (ARCH effect), with periods of elevated uncertainty followed by mean-reversion.

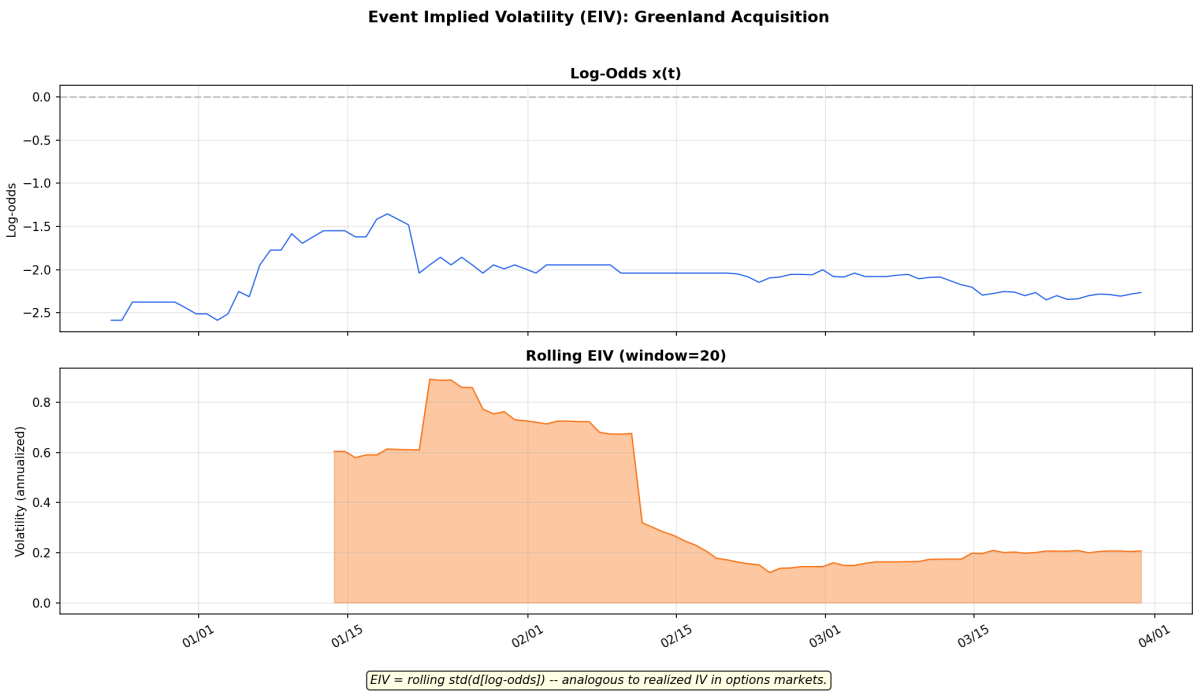

Figure 10: EIV for the Greenland acquisition contract (Polymarket, daily). The mid-January spike coincides with Trump administration statements; EIV decays as the market digests information, consistent with information arrival models (Easley and O'Hara, 1992).

EIV provides a descriptive tool for measuring the rate of probability revision—conceptually related to realized volatility measures in equity markets, though not directly analogous to the VIX, which is derived from option-implied volatilities. Combined with the Wang transform, it offers a two-dimensional characterization of prediction market contracts: λ captures the level of the pricing wedge, while σ_{EIV} captures the rate of information arrival.

Cross-contract empirical distribution. To move beyond case studies, I compute realized event volatility for all $N = 14{,}406$ contracts with ≥ 20 hourly price observations, using a rolling window of $w = 10$ observations and annualizing by $\sqrt{8{,}760}$. Table 20 reports the cross-sectional distribution. The median annualized realized volatility is $\hat{\sigma}_{\text{realized}} = 5.31$, with substantial right skew (mean $= 12.54$) and wide interquartile range $[1.52, 16.73]$. Crypto contracts exhibit the highest median volatility (11.99), followed by Tech (14.06) and Politics (8.64); Sports contracts cluster near the overall median (5.50). Volume has negligible correlation with realized volatility (Spearman $\rho = 0.015$, $p = 0.10$), suggesting that EIV captures information arrival intensity rather than liquidity effects.

An ARCH(1) Lagrange multiplier test on squared log-odds increments rejects the null of no volatility clustering at the 5% level for 22.4% of contracts (3,231 out of 14,406), confirming that volatility clustering—a hallmark of financial time series (Cont, 2001)—is present in a substantial fraction of prediction market contracts. The rejection rate varies by category: Crypto contracts show the highest clustering (35%), consistent with the speculative dynamics in digital asset

markets, while Sports contracts show the lowest (12%), consistent with information arriving in discrete bursts (game outcomes) rather than continuous flow.

Table 20: Cross-contract distribution of realized event volatility $\hat{\sigma}_{\text{realized}}$, computed as the median rolling standard deviation of hourly log-odds increments (annualized, window = 10). Contracts with ≥ 20 hourly observations.

Group	N	Median	IQR	Mean	ARCH(1) rej. rate
All contracts	14,406	5.31	[1.52, 16.73]	12.54	22%
By category					
Sports	3,731	5.50	[1.92, 17.70]	14.01	12%
Politics	603	8.64	[1.66, 15.84]	13.99	20%
Crypto	910	11.99	[0.00, 20.70]	14.56	35%
Tech	654	14.06	[0.59, 29.60]	19.24	18%
Other	8,508	4.39	[1.36, 14.61]	11.07	26%
By volume tier					
Low (<$500)	4,664	2.71	[0.00, 11.47]	10.37	23%
Med ($500–2K)	2,533	5.44	[2.14, 17.57]	14.47	19%
High ($2K–10K)	2,522	8.34	[2.33, 19.62]	14.81	21%
Very high (>$10K)	4,687	7.15	[2.35, 18.13]	12.45	24%

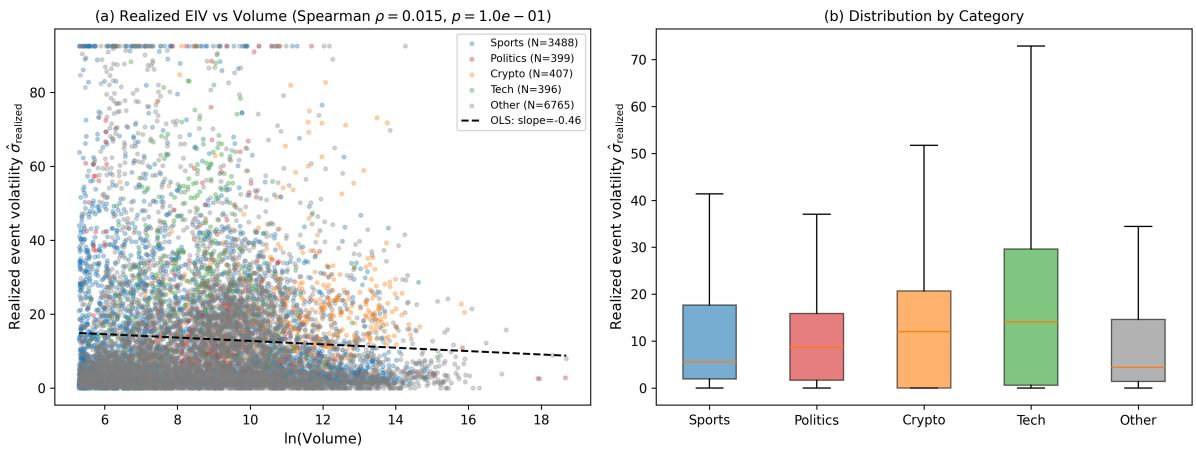

Figure 11: Cross-contract distribution of realized event volatility. Panel (a): scatter of $\hat{\sigma}_{\text{realized}}$ against ln(Volume), colored by event category; the OLS line (slope $= -0.46$) and Spearman rank correlation ($\rho = 0.015$, $p = 0.10$) confirm negligible volume–volatility association. Panel (b): box plots by category, showing Crypto and Tech contracts exhibit higher and more dispersed realized volatility than Sports or Politics.

8 Conclusion

This paper develops a structural pricing framework for prediction markets based on incomplete markets and maintained pricing-measure selection rules. The central observation is that binary event claims are generally not spanned by traded assets, so no-arbitrage determines a set of admissible pricing measures rather than a unique price. To organize prices within that incomplete-market set, I impose a latent Gaussian threshold model under the physical measure together with a one-parameter exponential tilt of the standardized latent factor. Under this maintained Gaussian family, the Wang transform arises exactly as a variance-preserving pricing-measure selection rule. The framework therefore provides a tractable structural benchmark for mapping between physical probabilities and observed market prices, and yields favorite-longshot bias as a corollary rather than as a separate behavioral primitive.

The empirical results support the framework across multiple dimensions. On Polymarket, the estimated selection parameter is positive and highly significant in both the baseline and expanded samples. Across 291,309 contracts from multiple platforms, the estimate remains positive on real-money venues and turns negative on the play-money platform Manifold. The wedge is larger in lower-liquidity and longer-duration markets, declines over contract life, and is robust to alternative opening-price constructions. An external benchmark using Elo probabilities for NBA contracts provides independent directional support for a positive wedge, though sharper benchmark data such as sportsbook closing lines would strengthen identification.

Limitations. The calibration recovers the composite selection parameter within the maintained Wang family; separately identifying its risk-bearing and behavioral components requires instruments or natural experiments. Candidates include regulatory shocks (Kalshi CFTC ruling), the time-varying pattern documented in Section 5.4 (which suggests market-microstructure rather than behavioral origins), and the cross-platform dimension (same event, different λ). The Manifold sign reversal (Section 6) provides suggestive evidence for the risk-bearing interpretation but is not a definitive causal identification. For the Kalshi cross-platform analysis, the available price data is the last traded price before settlement rather than the opening price used for Polymarket; this limits the interpretation of calibration improvements (ECE, Brier score) for Kalshi, though the $\hat{\lambda}$ point estimates remain valid as they reflect the systematic relationship between price levels and resolution rates. At extreme probabilities, the elevated $\hat{\lambda}$ may reflect confidence-related distortions rather than a purely risk-bearing interpretation of the estimated selection parameter; the extended model finds extremity is a significant predictor but cannot distinguish risk from misperception. The external benchmark analysis (Section 5.6), using an Elo rating model as an independent probability estimate for 51 NBA contracts, provides directional support for a positive pricing wedge ($\hat{\lambda} = 0.125$, $p = 0.014$), though the sample is small and the Elo model is noisier than sportsbook closing lines. Replication with commercial closing-line data would strengthen this identification.

To the best of my knowledge, no prior work has applied the Wang transform to prediction markets, and the empirical results—spanning 291,309 contracts across eight data sources, six plat-

forms, and eleven years—confirm that the framework captures economically meaningful structure in the data. The framework connects prediction market pricing to three established literatures: (i) distortion risk measures in insurance mathematics (Wang, 2000; Yaari, 1987); (ii) the interpretation of prediction market prices in economics (Manski, 2006; Wolfers and Zitzewitz, 2006); and (iii) incomplete market pricing in mathematical finance (Harrison and Pliska, 1983). By bridging these literatures, it provides practitioners with a principled, model-dependent pricing benchmark—conditional on the maintained Wang family and its identifying assumptions—and researchers with a testable structural framework for studying price-probability wedges in prediction markets.

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A Supplementary Tables

Table 21: Per-horizon separate MLEs (baseline calibration, 12-day subsample). These estimates are superseded by the stacked panel model (Section 5.4), which borrows strength across horizons and provides correct contract-clustered inference.

Lifetime %	N	$\hat{\lambda}_{MLE}$	SE	95% CI	p-value
5% (Opening)	2,460	0.176***	0.027	[0.123, 0.230]	7.1×10^{-11}
10%	2,409	0.158***	0.027	[0.104, 0.212]	8.3×10^{-9}
20%	2,316	0.110***	0.028	[0.055, 0.165]	8.8×10^{-5}
30%	2,236	0.082**	0.029	[0.026, 0.139]	0.004
40%	2,065	0.080**	0.030	[0.021, 0.138]	0.008
50% (Mid-life)	1,976	0.050	0.030	[−0.010, 0.110]	0.102
60%	1,902	0.052	0.031	[−0.008, 0.113]	0.091
70%	1,763	0.040	0.032	[−0.023, 0.103]	0.216
80%	1,662	0.031	0.033	[−0.035, 0.096]	0.357
90%	1,359	0.062	0.038	[−0.011, 0.136]	0.098

Table 22: One-stage hierarchical Wang MLE on Kalshi contracts. Dependent variable: binary outcome y_i. Model: $\Pr(y_i = 1) = \Phi(\Phi^{-1}(p_i^{\text{mkt}}) - X_i\beta)$. Fisher standard errors reported; sandwich robust SEs in brackets.

	Main spec		Year FE	
Variable	Coef.	SE	Coef.	SE
Constant	0.1400***	(0.0198)	0.3357**	(0.1102)
	[0.0188]		[0.1032]	
ln(Volume)	0.0384***	(0.0031)	0.0300***	(0.0032)
	[0.0029]		[0.0031]	
ln(Duration)	-0.1101***	(0.0032)	-0.1241***	(0.0035)
	[0.0030]		[0.0033]	
$\lvert p - 0.5 \rvert$	0.1917***	(0.0267)	0.1768***	(0.0268)
	[0.0261]		[0.0262]	
Year=2022			0.0725	(0.1115)
			[0.1045]	
Year=2023			0.0444	(0.1083)
			[0.1012]	
Year=2024			0.0028	(0.1077)
			[0.1005]	
Year=2025			-0.0847	(0.1079)
			[0.1007]	
Year=2026			-0.1272	(0.1070)
			[0.1001]	
N		200,226		200,226
Log-lik.		-75541.5		-75475.4
LR χ^2 (covariates)		1267.2		1399.4
Pseudo R^2		0.024828		0.025681

Table 23: Cross-platform category-level Wang parameter comparison. Each cell reports $\hat{\lambda}_{MLE}$ estimated separately by category. $\Delta\lambda = \hat{\lambda}_{\text{Kalshi}} - \hat{\lambda}_{\text{Polymarket}}$; positive values indicate higher risk loading on the less liquid platform.

	Polymarket			Kalshi			
Category	N	$\hat{\lambda}$	SE	N	$\hat{\lambda}$	SE	$\Delta\lambda$
Sports	973	0.070	0.042	122	0.573***	0.158	+0.503
Politics	71	0.054	0.157	750	0.400***	0.066	+0.346
Crypto	305	0.253	0.077	12,975	0.297***	0.015	+0.044
Tech	414	0.268	0.079	8,246	-0.022	0.017	-0.290
Other	692	0.282	0.051	178,133	0.178***	0.004	-0.104
Pooled	2,460	0.176	0.027	200,226	0.178***	0.004	+0.002

Table 24: Polymarket timing ladder: one-stage hierarchical Wang MLE estimated at different lifecycle pricing points. The Kalshi pooled estimate (using near-settlement prices) is shown in the last row for comparison. Robust (sandwich) standard errors in parentheses.

Timing point	N	$\hat{\beta}_0$	$\hat{\beta}_{\text{vol}}$	$\hat{\beta}_{\text{dur}}$	$\hat{\beta}_{\text{ext}}$
PM 5% (opening)	13,274	0.2599*** (0.0636)	-0.0716*** (0.0053)	0.1431*** (0.0122)	-0.4778*** (0
PM 20%	12,923	0.2097** (0.0646)	-0.0616*** (0.0053)	0.1248*** (0.0125)	-0.3617*** (0
PM 50% (mid-life)	12,645	0.1797** (0.0648)	-0.0541*** (0.0053)	0.1050*** (0.0125)	-0.1907* (0.0
PM 80% (late-life)	10,697	0.2884*** (0.0743)	-0.0467*** (0.0056)	0.0561*** (0.0148)	-0.2079* (0.1
Kalshi (near-settle.)	200,226	0.1400*** (0.0188)	0.0384*** (0.0029)	-0.1101*** (0.0030)	0.1917*** (0.
Sign comparison (Polymarket opening → Kalshi):					
		$+ \to +$ ✓	$- \to +$ ×	$+ \to -$ ×	$- \to +$

Table 25: Distance between Polymarket and Kalshi slope coefficients at each timing point. $\Delta\beta_j = \hat{\beta}_j^{\text{PM}} - \hat{\beta}_j^{\text{Kalshi}}$ for each covariate j. Decreasing distances indicate convergence toward the Kalshi coefficient pattern.

	Signed difference $\Delta\beta$			Distance		
Timing	$\Delta\beta_{\text{vol}}$	$\Delta\beta_{\text{dur}}$	$\Delta\beta_{\text{ext}}$	L_1	L_2	Reduction
PM 5% (opening)	-0.1101	+0.2532	-0.6695	1.0328	0.7242	—
PM 20%	-0.1000	+0.2349	-0.5535	0.8883	0.6095	14%
PM 50% (mid-life)	-0.0925	+0.2151	-0.3824	0.6900	0.4484	33%
PM 80% (late-life)	-0.0851	+0.1662	-0.3997	0.6510	0.4411	37%

A.1 External Benchmark Details

This appendix provides methodological details for the external benchmark validation reported in Section 5.6.

Elo model specification. I build Elo ratings from the complete 2025–26 NBA regular season (1,135 games, 30 teams) using a standard recursive update. The parameters are $K = 28$ (update factor) and home-court advantage = 42 Elo points, calibrated to the modern NBA home-win rate of approximately 56%. All teams are initialized at Elo = 1500; preseason and All-Star games are excluded.

Elo calibration. The model achieves a Brier score of 0.218 on the 1,135-game sample. Binned calibration shows reasonable alignment between predicted and realized home-win rates, with the strongest calibration in the 30–80% predicted probability range.

Sportsbook validation. To validate the Elo model against a sharper external benchmark, I compare its predictions to consensus moneyline odds from The Odds API for 13 contemporaneous NBA games. The consensus probability is the mean across nine bookmakers, with overround removed by normalization. The correlation between Elo and sportsbook probabilities is $r = 0.618$, with MAE = 0.166 and RMSE = 0.200. The moderate correlation confirms that the Elo model captures a substantial share of the variation in sharp-market pricing, while the residual noise inflates the contract-level $\hat{\lambda}_i$ estimates relative to what a sportsbook-based benchmark would yield.

Matching procedure. I identify Polymarket NBA game contracts by regex matching on the title format "Team A vs. Team B'', requiring both team names to normalize to one of 30 canonical NBA team names via an alias table. Each contract is matched to the closest Elo game involving the same team pair within a $[-1, +5]$-day window around the contract's creation date. Of 80 contracts passing team normalization, 51 are successfully matched; the 29 unmatched contracts reference games outside the matching window or future unplayed games.

Contract-level $\hat{\lambda}_i$. For each matched contract, the Wang parameter is computed as $\hat{\lambda}_i = \Phi^{-1}(p_i^{\text{Polymarket}}) - \Phi^{-1}(\hat{p}_i^{\text{Elo}})$, where $p_i^{\text{Polymarket}}$ is the opening price (first-5% midpoint) and $\hat{p}_i^{\text{Elo}}$ is the pre-game Elo probability assigned to the first-named team.

OLS regression. A cross-sectional regression of $\hat{\lambda}_i$ on ln(Volume) and price extremity $|p_i - 0.5|$ finds neither covariate significant at conventional levels ($N = 51$; Table 16, Panel B), though the negative extremity coefficient ($\hat{\beta} = -0.72$, $p = 0.094$) is suggestive of pricing wedge concentrated near even-odds contracts. The limited power reflects the small matched sample.